Further Maths Revision Notes

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Chapter 1

Core Pure Year I

Core Pure I Textbook (C1)

1.1 Complex Numbers I

1.1.1 The Imaginary Constant

Using only the real numbers, there is no number x that satisfies the equation $x^2+1=0$. To satisfy this polynomial, you must first define a square root of -1. This constant is called the imaginary constant i, defined such that $i^2=-1$.

Using the imaginary constant, complex numbers can be defined as $a+bi$, where $a, b \in \mathbb{R}$. The set of all numbers in the form $a+bi$ is called the set of complex numbers, written $\mathbb{C}$.

The functions $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ (sometimes written as $\mathfrak{R}(z)$ and $\mathfrak{I}(z)$) take the real part and imaginary part of a complex number respectively. For example,

$$\mathrm{Re}(a+bi) \equiv \mathfrak{R}(z) := a,\ a, b \in \mathbb{R} \tag{1.1}$$

$$\mathrm{Im}(a+bi) \equiv \mathfrak{I}(z) := bi,\ a, b \in \mathbb{R} \tag{1.2}$$

1.1.2 Complex Arithmetic

Complex numbers can be added and multiplied similarly to how binomials would be. Given two complex numbers $w := a+bi$ and $z := c+di$, where $a, b, c, d \in \mathbb{R}$, the following rules define complex addition and multiplication:

$$w+z = (a+bi)+(c+di) = (a+c)+(b+d)i \tag{1.3}$$

$$w \times z = (a+bi)(c+di) = (ac-bd)+(ad+bc)i \tag{1.4}$$

1.1.3 Complex Conjugates

The complex conjugate of a complex number of z, written as z^* is z with a flipped sign before the imaginary part.

$$(a+bi)^* := a-bi,\ a,b\in\mathbb{R} \tag{1.5}$$

From this definition, we can see that the sum and product of a complex number and its conjugate are both real. Let $z := a+bi$, where $a,b\in\mathbb{R}$.

$$z+z^* = 2a\in\mathbb{R} \tag{1.6}$$

$$zz^* = a^2+b^2\in\mathbb{R} \tag{1.7}$$

Using the multiplication rule in equation 1.7, a fraction of complex numbers can be 'realised' by multiplying by the complex conjugate of the denominator over itself. Realising the denominator means rewriting a fraction of complex numbers as a fraction with a real denominator.

$$\begin{aligned}\frac{a+bi}{c+di} &= \frac{a+bi}{c+di}\times\frac{c-di}{c-di}\\ &= \frac{(a+bi)(c+di)}{c^2+d^2}\\ &= \frac{(ac-bd)+(ad+bc)i}{c^2+d^2}\end{aligned} \tag{1.8}$$

1.1.4 Complex Roots of Polynomials

All nth degree polynomials (polynomials with a power of x up to x^n) have exactly n roots, counting complex roots and algebraic multiplicities. Therefore, if an equation does not cross $y=0$ on a Cartesian graph, it must have other, complex roots. The simplest example of this is the quadratic $x^2+1=0$.

$$x^2 = -1$$

$$x = \pm i$$

For quadratics, whenever the discriminant ($\Delta := b^2-4ac$ for a quadratic ax^2+bx+c) is less than 0, the quadratic has complex roots, since the quadratic formula (B.1) can be rewritten as

$$x = \frac{-b\pm\sqrt{\Delta}}{2a}$$

which would give a complex number whenever $\Delta<0$, since $\sqrt{x}$ is complex iff $x<0$ ($x\in\mathbb{R}$).

If a polynomial has a complex root α, it must also have the complex conjugate root α^*. For example, if a quadratic has a root $\alpha = 2+3i$, then the other root must be $\alpha^* = 2-3i$. For a cubic, there can only be one pair of roots, thus the third root cannot be a complex root, since it would have to have a fourth root to form a conjugate pair. Thus, the third root of a cubic must a real number.

1.1.5 Example Questions

Question 1.1.1. *From C1; question 1A/8*: Given that $z = a + bi$ and $w = a - bi$, $a, b \in \mathbb{R}$, show that:

a) $z + w$ is always real

b) $z - w$ is always imaginary

Answer 1.1.1:

a) $z+w$ can be expanded to $(a+bi)+(a-bi)$, which simplifies to $2a+bi-bi = 2a$. Since $a \in \mathbb{R}$, $2a \in \mathbb{R}$, thus $z + w \in \mathbb{R}$. □

b) $z-w$ can be expanded to $(a+bi)-(a-bi)$, which simplifies to $a-a+2bi = 2bi$. Since $b \in \mathbb{R}$, $2b \in \mathbb{R}$, thus $2bi$ is purely imaginary. □

Question 1.1.2. *From C1; question 1C/9*:

a) Given that $i^1 = i$ and $i^2 = -1$, write i^3 and i^4 in their simplest forms.

b) Write i^5, i^6, i^7 and i^8 in their simplest forms.

c) Write down the value of:

i) i^{100}

ii) i^{253}

iii) i^{301}

Answer 1.1.2:

a) Since $i^2 = -1$, i^3 must be $i^2 \times i = -1 \times i = \boxed{-i}$. Then, i^4 is $i^3 \times i = -i \times i = -(i^2) = -(-1) = \boxed{1}$.

b) i^5 is $i^4 \times i = 1 \times i = \boxed{i}$; i^6 is $i^5 \times i = \boxed{-1}$; $i^7 = i^6 \times i = \boxed{-i}$; $i^8 = i^7 \times i = \boxed{1}$.

c) From the pattern in parts a and b, it is clear that $i^{n+4k} = i^n$ for all $k \in \mathbb{Z}$, thus $i^n = i^{n \bmod 4}$.

i) $100 \mod 4 = 0$, so $i^{100} = i^0 = \boxed{1}$.

ii) $253 \mod 4 = 1$, so $i^{253} = i^1 = \boxed{i}$.

iii) $301 \mod 4 = 1$, so $i^{301} = i^1 = \boxed{i}$.

Question 1.1.3. *From C1; question 1C/Challenge*:

a) Expand $(a + bi)^2$

b) Hence, or otherwise, find $\sqrt{40 - 42i}$, giving your answer in the form $a-bi$, where a and b are positive integers.

Answer 1.1.3:

a) $(a + bi)^2 = \boxed{a^2 + 2abi - b^2}$.

b) Let $z = a - bi = \sqrt{40 - 42i}$. Thus, $z^2 = 40 - 42i = a^2 - 2abi - b^2$. This shows that $40 = a^2 - b^2$ and $-42 = -2ab$, since the real and imaginary terms must match up. Rearranging $-42 = 2ab$ in terms of a gives:

$$\begin{aligned} -42 &= -2ab \\ 21 &= ab \\ a &= \frac{21}{b} \end{aligned}$$

Which can then be substituted into $40 = a^2 + b^2$:

$$\begin{aligned} 40 &= (\frac{21}{b})^2 - b^2 \\ 40 &= \frac{441}{b^2} - b^2 \\ 40b^2 &= 441 - b^4 \end{aligned}$$

This is now a quadratic in terms of b^2, so we'll define $\beta = b^2$, and it can now be solved using the quadratic equation:

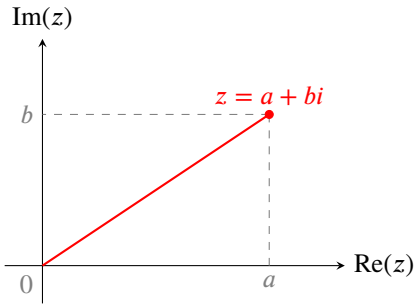
$$\begin{aligned} b^4 + 40b^2 - 441 &= 0 \\ \beta^2 + 40\beta - 441 &= 0 \\ \beta &= \frac{-40 \pm \sqrt{40^2 - 4(-441)}}{2} \\ &= \frac{-40 \pm \sqrt{3364}}{2} \\ &= \frac{-40 \pm 58}{2} \\ &= -20 \pm 29 \\ \beta = 9 \text{ or } \beta &= -49 \\ b &= \pm\sqrt{\beta} \end{aligned}$$

Thus, $b = 3$ or $b = -3$ or $b = 7i$ or $b = -7i$. Since the question stipulates that *"a and b are positive integers"*, b must be 3. Therefore, $a = \frac{21}{3} = 7$. So $\sqrt{40 - 42i} = \boxed{7 - 3i}$.

1.2 Argand Diagrams

1.2.1 Complex Numbers on Argand Diagrams

An Argand diagram can be thought of similarly to a Cartesian graph; complex numbers in the form $x + iy$ can be plotted like (x, y) can on a Cartesian graph. Instead of an x axis and a y axis, there are instead a real axis and an imaginary axis, used to plot the real part and imaginary part of complex numbers. An Argand diagram looks like this:



Where the point $z = a + bi$ is drawn where the point (a, b) would be drawn on a Cartesian graph.

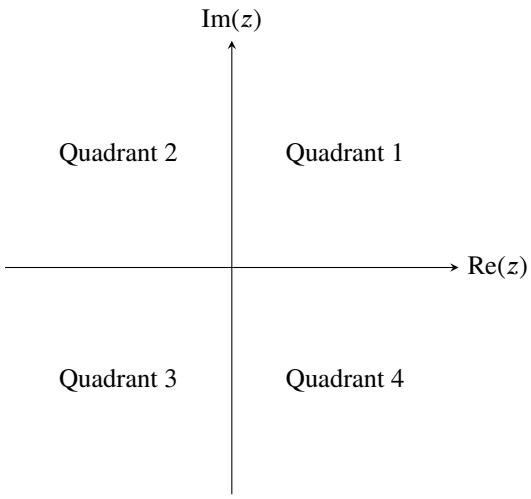
1.2.2 Modulus-Argument Form

The modulus-arguemnt form of a complex number is similar to the polar form of a point on a Cartesian graph. It defines the point using its distance from the origin, and the angle it makes with the positive half of the real axis, in radians. The modulus of z is written as $|z|$, which is calculated using Pythagoras' Theorem:

$$|z| = \sqrt{\mathrm{Re}^2(z) + \mathrm{Im}^2(z)} \tag{1.9}$$

where $\mathrm{Re}^2(z)$ and $\mathrm{Im}^2(z)$ mean $(\mathrm{Re}(z))^2$ and $(\mathrm{Im}(z))^2$ respectively. The angle that z makes with the positive real number line is called the argument, and is positive, going anti-clockwise if $\mathrm{Im}(z) \geq 0$, and negative, going clockwise if $\mathrm{Im}(z) < 0$. The argument of z is written as $\arg(z)$.

An Argand diagram is split into four quadrants, called quadrant 1, 2, 3 and 4:



For values of z in the first and fourth quadrants ($\text{Re}(z) > 0$):

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) \tag{1.10}$$

For values of z in the second quadrant:

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) + \pi \tag{1.11}$$

For values of z in the third quadrant:

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) - \pi \tag{1.12}$$

For a complex number $z = x + iy$ with modulus $|z| = r$ and argument $\arg z = \theta$, the following are true from right-angled trigonometry:

$$x = r\cos\theta \tag{1.13}$$
$$y = r\sin\theta \tag{1.14}$$

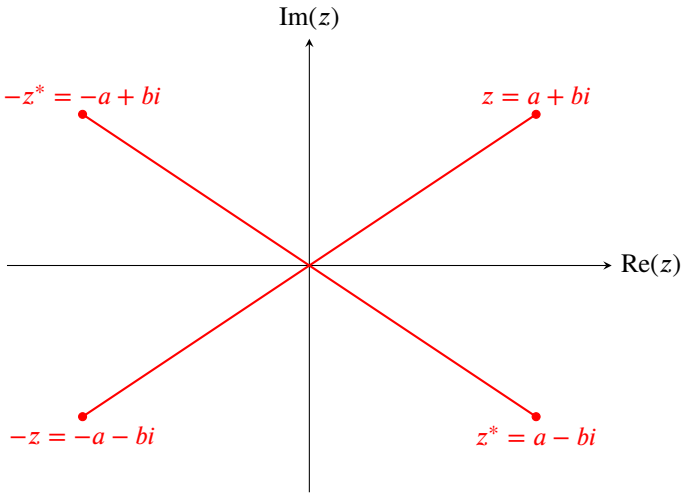
Thus, z can be written as:

$$z = r(\cos\theta + i\sin\theta) \tag{1.15}$$

To shorten this form, the function $\text{cis}\,\theta$ (short for **c**osine plus ***i*** **s**ine) can be used to simplify it down to:

$$z = r\,\text{cis}\,\theta \tag{1.16}$$

On an Argand diagram, the complex conjugate of a number is itself mirror in the line $\text{Im}(z) = 0$ (see subsection 1.2.3):



Using the fact that the negative of a complex number is itself rotated by $180° \equiv \pi$ radians, a complex number can be mirrored over the lines $\mathrm{Re}(z) = 0$, $\mathrm{Im}(z) = 0$, and $\mathrm{Re}(z) = \pm \mathrm{Im}(z)$.

Looking that this graph, it is clear to see that the argument of the complex conjugate of a complex number is negative that of the original complex number, thus:

$$\arg z = -\arg z^* \tag{1.17}$$

$$r \operatorname{cis} \theta = r \operatorname{cis}(-\theta) \tag{1.18}$$

Complex Numbers in the Form $r(\cos\theta - i\sin\theta)$

Complex numbers can be written in the form $r(\cos\theta - \sin\theta)$, however this is not generally helpful, so they should be converted to regular $r\operatorname{cis}\theta$ form. This can be done using the identities:

$$\sin\theta = -\sin(-\theta) \tag{1.19}$$

$$cos\theta = \cos(-\theta) \tag{1.20}$$

These identities show that sin is an odd function ($f(x) = -f(-x)$), and cos is an even function ($f(x) = f(-x)$). This means that sin has a rotational symmetry of 2 when rotated around the origin, and cos has a line of symmetry at $x = 0$.

From this, the $-\sin\theta$ term can be rewritten as $\sin(-\theta)$, and $\cos\theta$ can match with $\cos(-\theta)$, to make sure that the angle in the cosine and sine terms are the same. From there, that gives:

$$r(\cos\theta - i\sin\theta) = r\left(\cos(-\theta) + i\sin(-\theta)\right) = r\operatorname{cis}(-\theta) \tag{1.21}$$

And since have negated angles, that means that:

$$r\operatorname{cis}(-\theta) = (r\operatorname{cis}\theta)^* \tag{1.22}$$

1.2.3 Loci and Regions in Argand Diagrams

A locus (plural loci) is a set of points satisfying some criteria. For example, the locus of points such that, for all points z, $|z| = 1$ forms a circle on the complex plane with a radius of 1, since the set of points $|z| = 1$ is the set of points 1 unit away from the centre, by the definition of the modulus.

Circles in the Complex Plane

Generalising the $|z| = 1$ locus:

Theorem 1.2.1. *A circle of any radius r can be made with the locus $|z| = r$.*

Proof. Given the equation of a circle centred on $(0, 0)$ in the Cartesian plane:

$$x^2 + y^2 = r^2 \tag{1.23}$$

We can define $z = x + iy$, which means that $|z| = \sqrt{x^2 + y^2}$. Thus, squaring both sides of the equation $|z| = r$ gives:

$$x^2 + y^2 = r^2 \tag{1.24}$$

Which is the same as the Cartesian equation for a circle. □

The circle can then be moved to be centred on a point $a + bi$, similarly to how a circle in the Cartesian plane can be moved to a point (a, b):

Theorem 1.2.2. *A circle centred on $a + bi$ in the complex plane has the equation:*

$$|z - (a + bi)| = r \tag{1.25}$$

Proof. Using the Cartesian equation for a circle centred on (a, b):

$$(x - a)^2 + (y - b)^2 = r^2 \tag{1.26}$$

We can define $z = x + iy$. Thus, the modulus of $z - (a + bi)$ is:

$$\begin{aligned} |z - (a + bi)| &= |(x - a) + (y - b)i| \\ &= \sqrt{(x - a)^2 + (y - b)^2} \end{aligned}$$

Thus, squaring each side of the equation $|z - (a + bi)| = r$ gives:

$$(x - a)^2 + (y - b)^2 = r^2 \tag{1.27}$$

Which is the Cartesian equation, thus they are the same. □

On an Argand diagram, this is what the locus $|z - (a + bi)| = r$ looks like:

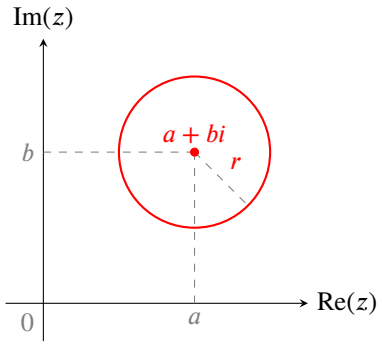


Figure 1.1: The locus $|z-(a+bi)|=r$ on an Argand diagram

Lines in the Complex Plane

Similar how you can make a locus from $|z| = r$, you can also make a locus from $\arg z = \theta$.

Theorem 1.2.3. *The locus* $\arg z = \theta$ *is a ray*[1] *from 0 at an angle of* θ.

Proof. First, we will turn the equation $\arg z = \theta$ into its Cartesian form using:

$$\theta = \arctan\left(\frac{y}{x}\right) \tag{1.28}$$

This means that:

$$\tan\theta = \frac{y}{x}$$
$$y = x\tan\theta$$

This is the Cartesian equation for a line rotated from $y = 0$ by an angle θ. However, the locus cannot be a line; since two points opposite each other on an Argand diagram have different arguments. For example, the points $\operatorname{cis}\left(\frac{\pi}{4}\right)$ and $-\operatorname{cis}\left(\frac{\pi}{4}\right)$ have arguments of $\frac{\pi}{4}$ and $-\frac{3\pi}{4}$ respectively. This means that the locus cannot cross 0. Since $\arg z$ is undefined for $z = 0$, the locus also cannot contain 0. This leaves a ray from 0 (not including 0) to infinity with the angle θ from the positive real line. □

Similar to how the circlular locus could be moved by changing z to $z-(a+bi)$, the ray locus can also be moved by subtracting $(a+bi)$ from z.

Theorem 1.2.4. *The locus* $\arg(z-(a+bi)) = \theta$ *is a ray with angle* θ *from the point* $a+bi$.

[1] A ray is a similar to a line, except it has one end point, and extends infinitely in the other direction. See figure 1.2.3 for an example.

Proof. We will define $z = x + iy$, and thus we can convert $\arg(z - (a + bi)) = \theta$ to:

$$\begin{aligned} \theta &= \arctan\left(\frac{y-b}{x-a}\right) \\ \tan\theta &= \frac{y-b}{x-a} \\ y - b &= (x-a)\tan\theta \end{aligned}$$

Which is the Cartesian equation for a line through a point (a, b). Using the same logic as in the proof for theorem 1.2.3 – the points on either side of 0 cannot have the same argument, thus after moving to $a+bi$, the line is still only a ray from (and not including) $a + bi$, at the angle θ. □

On an Argand diagram, this is what the locus $\arg(z - (a + bi)) = \theta$ looks like:

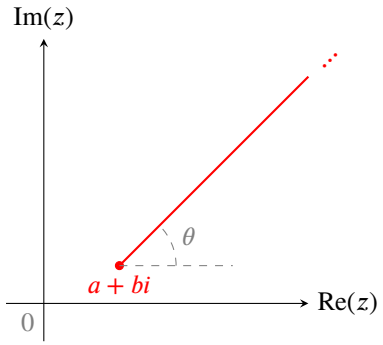


Figure 1.2: The locus $\arg(z - (a + bi)) = \theta$ on an Argand diagram

1.2.4 Example Questions

1.3 Series I

1.4 Roots of Polynomials

1.4.1 Sums of Roots

The roots of polynomials can be related using Vieta's formulae (see section B.1.1). The roots of a general polynomial are written as α, β, γ, δ, etc., and the coefficients are written as a, b, c, d, e, etc., where a is the coefficient of the greatest power of the polynomial, b is the coefficient of the next greatest power, etc. The sum of all roots is then written as $\sum \alpha \equiv \alpha + \beta + \cdots$. Multiple roots can be multiplied together and then summed, for example the sum of all pairs of roots ($\alpha\beta + \alpha\gamma + \cdots$) would be written as $\sum \alpha\beta$. This would be the sum of *all* possible pairs of roots.

Vieta's formulae for quadratics ($ax^2 + bx + c$ with roots α, β) are:

$$\sum \alpha = -\frac{b}{a} \tag{1.29}$$

$$\alpha\beta = \frac{c}{a} \tag{1.30}$$

Vieta's formulae for cubics ($ax^3 + bx^2 + cx + d$ with roots α, β, γ) are:

$$\sum \alpha = -\frac{b}{a} \tag{1.31}$$

$$\sum \alpha\beta = \frac{c}{a} \tag{1.32}$$

$$\alpha\beta\gamma = -\frac{d}{a} \tag{1.33}$$

Vieta's formulae for quartics ($ax^4 + bx^3 + cx^2 + dx + e$ with roots α, β, γ, δ) are:

$$\sum \alpha = -\frac{b}{a} \tag{1.34}$$

$$\sum \alpha\beta = \frac{c}{a} \tag{1.35}$$

$$\sum \alpha\beta\gamma = -\frac{d}{a} \tag{1.36}$$

$$\alpha\beta\gamma\delta = \frac{e}{a} \tag{1.37}$$

More complex equations can be also formed using Vieta's formulae, by rearranging the formulae into other forms.

1.4.2 Other Forms of Vieta's Formulae

Theorem 1.4.1 (Sum of Reciprocals of Roots)**.** *For an n-degree polynomial (where $2 \leq n \in \mathbb{N}$), the sum of the reciprocal of all n roots is the sum of all combinations of products of $n-1$ root divided by the product of all roots.*

$$\sum^{n} \frac{1}{\alpha} = \frac{\sum \prod^{n-1} \alpha}{\prod \alpha} \tag{1.38}$$

For a quadratic ($n = 2$), this looks like:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\sum \alpha}{\alpha\beta} = \frac{\alpha + \beta}{\alpha\beta} \tag{1.39}$$

For a cubic ($n = 3$), this looks like:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\sum \alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \tag{1.40}$$

The numerator and denominator can then be simplified using Vieta's formulae.

Proof. For the simplest case, $n = 2$:

$$\begin{aligned}\frac{1}{\alpha} + \frac{1}{\beta} &= \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} \\ &= \frac{\alpha + \beta}{\alpha\beta}\end{aligned}$$

For the general case ($n \geq 2$), the sum of roots can be rewritten using common denominators. For a set of numbers S, the product of all numbers ($\prod S$) must be a common denominator, since dividing it by any $s \in S$ would give $\prod(S \setminus \{s\})$, meaning that it is divisible by s for any $s \in S$ ($s \mid \prod S \ \forall \ s \in S$). Thus, considering that we need a common denominator for all of the roots of the polynomial, we can write the common denominator as $\prod \alpha$ (the product of all roots). Now, when each reciprocal root $\left(\frac{1}{\alpha}\right)$ is multiplied on the top and bottom by $\prod \alpha$:

$$\frac{1}{\alpha} = \frac{\prod \alpha}{\alpha \prod \alpha} = \frac{\alpha\beta\gamma\cdots}{\alpha^2\beta\gamma\cdots} = \frac{\beta\gamma\cdots}{\alpha\beta\gamma\cdots} \tag{1.41}$$

$$\frac{1}{\beta} = \frac{\prod \alpha}{\beta \prod \alpha} = \frac{\alpha\beta\gamma\cdots}{\alpha\beta^2\gamma\cdots} = \frac{\alpha\gamma\cdots}{\alpha\beta\gamma\cdots} \tag{1.42}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \cdots = \frac{(\beta\gamma\cdots) + (\alpha\gamma\cdots) + \cdots}{\alpha\beta\gamma\cdots} \tag{1.43}$$

The root already on the denominator is now on the denominator twice (once from it already being there, twice from multiplying by $\prod \alpha$), and once on the numerator (from $\prod \alpha$), so it can cancel. This will leave the sum of all products of $n-1$ roots ($\sum \prod^{n-1} \alpha$), since the set of all products of $n-1$ roots is the product of all roots except α, the product of all roots except β, etc., which is exactly what is now in the numerator. □

Theorem 1.4.2 (Product of Roots to the Power n)**.** *For some n, the product of all roots to the power n is:*

$$\prod(\alpha^n) = \left(\prod \alpha\right)^n \tag{1.44}$$

Proof. First, we'll expand $\prod(\alpha^n)$ into:

$$\begin{aligned}\prod(\alpha^n) &= \alpha^n\beta^n\gamma^n\cdots \\ &= \underbrace{\alpha\times\alpha\times\cdots}_{n}\times\underbrace{\beta\times\beta\times\cdots}_{n}\times\underbrace{\gamma\times\gamma\times\cdots}_{n}\end{aligned}$$

This can then be rearranged by grouping one root from each α^n into a product, which can be simplified using th fact that $\prod\alpha \equiv \alpha\beta\gamma\cdots$:

$$\prod(\alpha^n) = \underbrace{(\alpha\beta\gamma\cdots)\times(\alpha\beta\gamma\cdots)\times\cdots}_{n} = \left(\prod\alpha\right)^n$$

□

Theorem 1.4.3 (Sum of Squares of Roots)**.** *The sum of squares of roots of a polynomial is:*

$$\sum(\alpha^2) = \left(\sum\alpha\right)^2 - 2\sum\alpha\beta \tag{1.45}$$

Proof. We will begin by expanding the expression $\left(\sum\alpha\right)^2$:

$$\left(\sum\alpha\right)^2 = (\alpha+\beta+\gamma+\cdots)(\alpha+\beta+\gamma+\cdots) \tag{1.46}$$

This can be expanded using a table with each root across the top and side:

$+$	α	β	γ	$\cdots$
α	α^2	$\alpha\beta$	$\alpha\gamma$	$\cdots$
β	$\alpha\beta$	β^2	$\beta\gamma$	$\cdots$
γ	$\alpha\gamma$	$\beta\gamma$	γ^2	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

From this table, it is clear that the sum of the diagonal is the sum of squares of roots ($\sum(\alpha^2)$), and the upper and lower triangles are identical, as well as being the set of all pairs of roots. Thus, the sum of elements in the table is:

$$\sum(\alpha^2) + 2\sum\alpha\beta \tag{1.47}$$

Which we have already established is equal to $\left(\sum\alpha\right)^2$, thus the equation can be rearranged to get:

$$\begin{aligned}\left(\sum\alpha\right)^2 &= \sum(\alpha^2) + 2\sum\alpha\beta \\ \sum(\alpha^2) &= \left(\sum\alpha\right)^2 - 2\sum\alpha\beta\end{aligned} \tag{1.48}$$

□

Theorem 1.4.4 (Sum of Cubes of Quadratic Roots)**.** *For a quadratic with roots α and β, the sum of cubes of roots is:*

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \tag{1.49}$$

Which can also be written:

$$\sum (\alpha^3) = \left(\sum \alpha\right)^3 - 3\alpha\beta \left(\sum \alpha\right) \tag{1.50}$$

Proof. We will begin by expanding $(\alpha + \beta)^3$:

$$\begin{aligned}(\alpha + \beta)^3 &= (\alpha + \beta)(\alpha + \beta)(\alpha + \beta) \\ &= (\alpha^2 + 2\alpha\beta + \beta^2)(\alpha + \beta) \\ &= \alpha^3 + \alpha^2\beta + 2\alpha^2\beta + 2\alpha\beta^2 + \alpha\beta^2 + \beta^3 \\ &= \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3\end{aligned}$$

This can then be rearrange by taking the $\alpha^3 + \beta^3$ off to the side, then factoring $3\alpha^2\beta + 3\alpha\beta^2$:

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta) \tag{1.51}$$

This can then be rearranged to give the final result of the proof:

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \tag{1.52}$$

□

As a more general version of theorem 1.4.4, the sum of cubes of any n-degree polynomials can be written as:

$$\sum (\alpha^3) = (\alpha)^3 - 3\left(\sum \alpha\right)\left(\sum \alpha\beta\right) + 3\left(\sum \alpha\beta\gamma\right) \tag{1.53}$$

For a quadratic, the $3\sum \alpha\beta\gamma$ term is 0 because there is no γ root.

1.4.3 Linear Transformations of Roots

1.5 Volumes of Revolution

1.6 Matrices

1.7 Linear Transformations

1.8 Proof by Induction

1.9 Vectors

Chapter 2

Core Pure Year II

Core Pure II Textbook (C2)

2.1 Complex Numbers II

2.2 Series II

2.3 Methods in Calculus

2.4 Polar Coordinates

2.5 Hyperbolic Function

2.6 Methods in Differential Equations

2.7 Modelling with Differential Equations

Chapter 3

Decision Year I

Decision I Textbook (D1)

3.1 Algorithms

3.1.1 Basic Algorithms

An algorithm is a set of instructions that tell a person or computer how to do a task. Algorithms can be represented in different ways. This is the 'happy algorithm' written as a set of steps of instructions and a flow chart:

1. Plain language steps (or pseudocode):
    - Write down an integer
    - Square its digits and find the sum of the squares
    - Continue with this number
    - Repeat until this number is 1 (the number is then considered 'happy') or until the algorithm gets trapped in a cycle (in which case it is considered 'unhappy')
2. Flow Charts (see section 3.1.1):

Trace Tables

Trace tables are used to track vales of different variables in an algorithm. For example, for the 'happy algorithm' (see trace table **TODO: ADD FIG REF**) could be traced for an input of 12:

Input number	Sum of square of digits		Output
12	5	$(1^2 + 2^2 = 5)$	-
5	25	$(5^2 = 25)$	-
25	29	$(2^2 + 5^2 = 29)$	-
29	85	$(2^2 + 9^2 = 85)$	-
85	89	$(8^2 + 5^2 = 89)$	-
89	145	$(8^2 + 9^2 = 145)$	-
145	42	$(1^2 + 4^2 + 5^2 = 42)$	-
42	20	$(4^2 + 2^2 = 20)$	-
20	4	$(2^2 + 0^2 = 4)$	-
4	16	$(4^2 = 16)$	-
16	37	$(1^2 + 6^2 = 37)$	-
37	58	$(3^2 + 7^2 = 58)$	-
58	89	$(5^2 + 8^2 = 89)$	Algorithm has entered a loop; 12 is unhappy.

A trace table for an algorithm can be created by following the algorithm through, recording each time each variable changes, in order. The trace table should be written in order, e.g. if there are variables x, y, z and x changes, then y changes, then x changes again before z changes, the change in z would be recorded after the second change in x, not in the top row of the z column.

Flow Charts

Flow charts are a way of writing algorithms, made up of boxes containing instructions. Each shape of box has a different meaning:

1. Rounded box (start/end): The start/end boxes, also known as terminators, are put at the beginning and end of an algorithm.

2. Rectangle (instruction): The instruction box tell the person reading the algorithm what steps have to be taken, for example a sorting algorithm such as bubble sort (see section 3.1.2) would have an instruction box telling the person reading the algorithm to swap two elements.

3. Diamond (decision): A decision box splits the algorithm's flow into multiple different directions based on a boolean decision. The decision will evaluate a statement, e.g. $a > b$?, and if it is true, it will go to one place to continue executing instructions, and if it is false, it will go somewhere else to continue executing different instructions. The question in a decision box is always phrased as '[boolean statement]?', e.g. '$x = 3$?', or '$\phi \wedge \psi$?'. As an example, bubble sort (see section 3.1.2) would have a decision similar to list$[i]$ > list$[i + 1]$? and if it is true, it will swap the two items, otherwise it will continue to the next step.

Combining these boxes can allow flow charts to represent more compex programming features, for example loops (flow arrows can connect boxes further on through the program flow to ones from earlier, and decisions can decide to break out of the loop and continue elsewhere).

3.1.2 Bubble Sort

Description

Bubble sort is an algorithm that sorts an unsorted list of data (a sorting algorithm). It passes through the list, checking each pair of items to see if they need to be swapped to be in order. For example, if it were sorting a list of numbers into numerical order and it found the pair 2, 1, it would swap them. This is an example of a single bubble sort pass, with each comparison marked:

```
[3  5]  8   2   5   6   3
 3 [5  8]  2   5   6   3
 3  5 [8  2]  5   6   3
 3  5  2 [8  5]  6   3
 3  5  2  5 [8  6]  3
 3  5  2  5  6 [8  3]
 3  5  2  5  6  3  8
```

Now, the 8 is in the right place, and the bubble sort will do another pass, ignoring the 8, since it must be in the right place after 1 pass, since it is the largest number. In general, after n passes, the last n items are sorted. Bubble sort gets its name from this property of the largets elements 'bubbling' to the top (end) of the list. If the algorithm does a pass without swapping any items, it must have already finished sorting the list, so it will return early.

Example of bubble sort:

Pass 1:	3	5	8	2	5	6	3
Pass 2:	3	5	2	5	6	3	**8**
Pass 3:	3	2	5	5	3	**6**	**8**
Pass 4:	2	3	5	3	**5**	**6**	**8**
Pass 6:	2	3	3	**5**	**5**	**6**	**8**
Pass 7:	2	3	**3**	**5**	**5**	**6**	**8**

No more swaps; sort complete.

Notice that pass 6 is the same as pass 7, as only on pass 7 has the algorithm not performed any swaps.

Order

Theorem 3.1.1. *Bubble sort has an order of* $\mathcal{O}(n)$

Proof. Let n be the number of elements in the list to be sorted. This means that (in the worst-case scenario) the bubble sort will have to do $i - 1$ pass to guarantee that every item has been sorted (since after the ith pass, i items have been sorted, $n - i$ items will be left to sort, the ith item must also be sorted). In the ith pass, $i - 1$ items will have been sorted, so the total number of comparisons will be:

$$\sum_{i=1}^{n}(n-i) = \frac{n(n-1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$$

Taking the fastest-growing term of $\frac{1}{2}n^2 + \frac{1}{2}n$ (the $\frac{1}{2}n^2$ term) and discarding the constant coefficient ($\frac{1}{2}$), this gives n^2, thus the order of the bubble sort algorithm is $\mathcal{O}\left(n^2\right)$. □

Implementation

In pseudocode, this is an example implementation of a bubble sort algorithm:

```
// Sorts a list using bubble sort; returns sorted list
// Example:
//     Input:  [3, 2, 7, 3, 5]
//     Output: [2, 3, 3, 5, 7]
//     Explanation: The unsorted input is sorted so that the
//                  numbers are returned in order
FUNC bubble_sort(list):            // Takes an unsorted list as
    VAR n = list.UPPER_BOUND       // input

    FOR (i = 0 TO n):                       // Goes through each
        VAR hasSwapped = FALSE              // pass
        FOR (j = 0 TO n - i):               // Goes through each
                                            // value in the list
            IF (list[i] > list[j + 1]):     // Checks if items need
                hasSwapped = TRUE           // swapping
                VAR temp = list[j + 1]      // Swaps items
                list[j + 1] = list[i]
                list[i] = temp
            END IF
        END FOR

        IF (NOT hasSwapped):   // If it doesn't find any items to
                               // swap, the list must be sorted
            OUTPUT("No more swaps; sort complete.")
            RETURN list        // Return the sorted list early,
        END IF                 // since there are no more swaps
    END FOR

    RETURN list                // Returns sorted list
END FUNC
```

3.1.3 Quick Sort

Description

A quick sort is a recursive sorting algorithm. It begins by choosing a 'pivot'; this is usually in the middle of the list (the $\lfloor \frac{n}{2} \rfloor$th element), but can be anywhere. Then, it splits the list into two smaller sub-lists, the left and the right. The left sub-list will contain all elements less than or equal to the pivot, and the right list will contain all elements greater than the pivot. The pivot is then left in the middle and is now sorted. Then, each sub-list with a length greater than 1 is sorted using quick sort, hence quicksort is recursive.

This is an example of quick sort (the pivot will be in brackets, and then sorted elements will be bold):

Pass 1:	3	5	8	(2)	5	6	3
Pass 2:	**2**	3	5	8	(5)	6	3
Pass 3:	**2**	3	(3)	5	**5**	8	(6)
Pass 4:	**2**	3	(3)	5	**5**	**6**	8

All sub-lists contain only 1 item; sort complete.

Order

Theorem 3.1.2. *Quick sort has an order of* $\mathcal{O}\left(n^2\right)$.

Proof. The worst case scenario for quicksort is where the pivot is consistantly the maximum/minimum pivot (i.e. `item[0]` or `item[n]`), leaving the sub-lists with lengths 0 and $n-1$. This means that the time taken to sort the list is the time taken for the nth pass plus the time taken to sort $n-1$ items. Therefore, sorting n items will take the time required to sort $n-1$ items + the time taken to sort $n-2$ items, to the time taken to sort 1 item. This can be represented as the sum:

$$T = \sum_{x=1}^{n} x = \frac{n(n+1)}{2} \tag{3.1}$$

Simplifying and removing constant coefficients, we get a time complexity of $\mathcal{O}\left(n^2\right)$. □

Implementation

In pseudocode, this is an implementation of quick sort using a pivot in the middle:

```
// Sorts list using quick sort; returns sorted list
// Example:
//     Input:  [3, 2, 7, 3, 5]
//     Output: [2, 3, 3, 5, 7]
//     Explanation: The unsorted list is sorted so that the
//                  numbers are returned in order.
FUNC quick_sort(list):              // Takes an unsorted list as
```

```
    VAR n = list.UPPER_BOUND        // input


    IF (n == 1):                    // Base case: n = 1
        RETURN list
    ELSE                            // Recursive case: n > 1
        VAR pivot = FLOOR(n / 2)    // Find the pivot index
                                    // The item at the pivot
                                    // index is list[pivot]

        VAR leftSublist  = []       // Initialise the sublists to
        VAR rightSublist = []       // sort items into

        FOR EACH item IN list:          // Sort each item into a
            IF (item < list[pivot]):    // sublist
                leftSublist.PUSH(item)
            ELSE
                rightSublist.PUSH(item)
            END IF
        END FOR

        VAR newList = []
        newList.JOIN(leftSublist)   // Join the left sublist onto
                                    // the new list
        newList.PUSH(list[pivot])   // Add the pivot item onto
                                    // the new list
        newList.JOIN(rightSublist)  // Join the right sublist
                                    // onto the new list

        RETURN quick_sort(newList)  // Sort and return the new
                                    // list
    END IF
END FUNC
```

3.1.4 Bin Packing Algorithms

A bin packing algorithm gives a way to fit a set of items with sizes into 'bins' with sizes. In abstracted terms, an item is a number, and a bin has a size – also a number. The set S can fit into a bin with size n if and only if $\sum S \leq n$.

Bin questions often use a set of bins of equal size, and ask how many are required.

Finding the fewest number of bins required for a given set of bins and a given set of values is an NP-hard problem, meaning that there exists an algorithm to find the fewest number of bins with an order of $\mathcal{O}((\,) n^k)$, $k \in \mathbb{R}^+$ if and only if P = NP.

To find a greatest lower bound for how many bins are required (for a set of bins where all bins are the same size), take the sum of all items, and divide by the size of a bin. This should be rounded up; if you need 2.5 bins, you actually need 3 to account for the extra 0.5. This gives the formula:

$$\text{LB} = \frac{\sum S}{\text{size}\,(B)} \tag{3.2}$$

First-Fit Bin Packing

Description The first-fit bin packing algorithm goes through each item in the list of items, and puts it into the first bin it finds with space for that item.

Order

Theorem 3.1.3. *The first-fit bin packing algorithm has an order of $\mathcal{O}(n \times k)$ for n items and k bins.*

Proof. The algorithm must iterate through every item to see if it will fit into a bin. The time taken per item scales by the amount of bins that there are. Assuming the worst case, i.e. only the last bin fits each item, the algorithm will have to check k bins, n times for n items, thus the order is $\mathcal{O}(n \times k)$. □

Implementation In pseudocode, this is an implementation of first-fit bin packing:

```
// Takes a list of items (REAL) and a list of bins
// (BIN(OF REAL)), then packs the items into the bins using the
// first-fit algorithm and returns the list of bins packed with
// items
// It will return an empty array if the items cannot be packed
// into the bins given
// Example:
//     Input:  items=[3, 2, 7, 3, 5]
//             bins=[BIN(size=10), BIN(size=5), BIN(size=10)]
//     Output: [BIN(items=[3, 2, 3]),
//              BIN(items=[5]),
//              BIN(items=[7])]
FUNC pack_first_fit(items, bins):
```

```
    FOR EACH item IN items:
        FOR EACH bin IN bins:           // Search all bins until
            IF (NOT bin.IS_FULL):       // Check that bin is not
                                        // full
                bin.items.PUSH(item)    // Add item to bin
                CONTINUE    // Continue to next item
                            // This skips the error return later
            END IF
        END FOR

        RETURN []           // Bins cannot be packed; return
                            // empty array
    END FOR
END FUNC
```

First-Fit Decreasing Bin Packing

Description First fit decreasing is like first-fit, but instead of iterating through the list of items normally, it sorts them into descending order instead, allowing the biggest items to fill up the space they need, then making the smaller items fit around them.

Order Since first-fir descending bin packing requires sorting the list, the order will depend on the order of the sorting algorithm.

Theorem 3.1.4. *Assuming the algorithm uses quick sort (order $\mathcal{O}\left(n^2\right)$), the order of the first-fit descending bin packing algorithm is $\mathcal{O}\left(n(n+k)\right)$*

Proof. The time taken for the first-fit decreasing bin packing algorithm is the time taken to sort the list, then the time taken to pack the bins. This means that the order will be $\mathcal{O}\left(n^2 + n \times k\right)$, which simplifies to $\mathcal{O}\left(n(n+k)\right)$. □

Implementation

```
// Takes a list of items (REAL) and a list of bins
// (BIN(OF REAL)), then packs the items into the bins using the
// first-fit decreasing algorithm and returns the list of bins
// packed with items
// It will return an empty array if the items cannot be packed
// into the bins given
// Example:
//     Input:  items=[3, 2, 7, 3, 5]
//             bins=[BIN(size=10), BIN(size=5), BIN(size=10)]
//     Output: [BIN(items=[7]),
//              BIN(items=[5, 3, 2]),
//              BIN(items=[2])]
FUNC pack_first_fit_decreasing(items, bins):
```

```
    items = reverse_sort(items)     // Sort the items in reverse

    // Regular first-fit bin packing algorithm:
    FOR EACH item IN items:
        FOR EACH bin IN bins:           // Search all bins until
            IF (NOT bin.IS_FULL):       // Check that bin is not
                                        // full
                bin.items.PUSH(item)    // Add item to bin
                CONTINUE    // Continue to next item
                            // This skips the error return later
            END IF
        END FOR

        RETURN []           // Bins cannot be packed; return
                            // empty array
    END FOR

END FUNC
```

Full-Bin Bin Packing

Full-bin bin packing involves using observation to fill bins using combinations of items that sum to the maximum size of a bin. This means that it cannot be easily by computers, and can be hard to do manually, since if there are a lot of items and a lot of bins, it can be hard to find sums that fill a bin.

3.1.5 Order of Algorithms

The order of an algorithm is how long it takes an algorithm to finish, based on the size of its input n, in the worst-case scenario. This is also called its time complexity, and is written as $\mathcal{O}((\,) f(n))$, where $f(n)$ is a function of the algorithm's input size.

For example, the linear search algorithm looks through each element in a list one by one, checking to try to find a specific item. The worst case for linear search is if the item being looked for is either not in the list, or is the final item in the list, meaning that the algorithm has to check all n elements. This means that linear search has an order of $\mathcal{O}(n)$.

The order of an algorithm can be used to find the time taken for an algorithm to finish based on other recorded times. For example, if an $\mathcal{O}(n)$ algorithm takes 5 ms to finish with 100 items, it will take 10 ms to finish with 200 items.

Common orders are $\mathcal{O}(1)$ (constant time), $\mathcal{O}(n)$ (linear time), $\mathcal{O}\left(n^2\right)$ (quadratic time), and $\mathcal{O}\left(n^3\right)$ (cubic time). Quadratic, cubic, etc. time are also called polynomial time ($\mathcal{O}\left(n^k\right),\ k \in \mathbb{R} \setminus \{0\}$). An algorithm is considered more efficient if its order scales slower with a bigger input. For example, if an algorithm runs in constant time ($\mathcal{O}(1)$), then that is considered very good, since with an arbitrarily big input, it will still run in the same amount of time. On the other hand, an algorithm with an order of $\mathcal{O}(2^x)$ (exponential time) would be considered inefficient and slow, since it scales very quickly as the input grows, making it very slow for large inputs.

3.2 Graphs and Networks

Graphs, (AKA networks) are sets of vertices (AKA nodes) that are connected by edges (AKA arcs). This is an example of a graph, with vertices A, B, C, D, and edges connecting A-B, B-C, A-C, A-D.

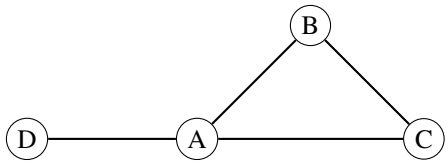


Figure 3.1: A simple graph

It does not matter what shape a graph is drawn in, as long as all of the vertices still exist and edges still connect the same vertices. This means that graph 3.1 is the same as this graph:

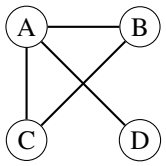


Figure 3.2: A graph that is the same as graph 3.1, but redrawn differently

Two graphs that share the same structure, even if the vertices are different (as long as there exists a bijection between the two sets of vertices). This means that 3.1 is isomorphic to 3.2. This also means that graph 3.2 and the following graph are isomorphic, even though the vertices are technically different[1]:

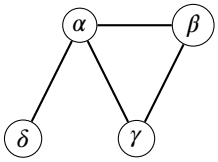


Figure 3.3: Another isomorphism of 3.1

Vertices have a property called 'degree', which means how many edges connect to that vertex. For example, in figure 3.1, vertex A has a degree of 3, because it is connected to B, C and D, where as vertex D has a degree of 1, because it is only connected to vertex A.

[1] A bijection between the vertices of the graphs could map A $\mapsto \alpha$, B $\mapsto \beta$, C $\mapsto \gamma$ and D $\mapsto \delta$, which preseves the structure of the edges.

The formal definition of a graph is an ordered pair of sets; one of vertices and edges. Each direction of an edge is represented by an ordered pair of vertices. An edge that connects both directions would be represented by both (A, B) and (B, A) being in the set of edges.

For graph 3.1, its definition in terms of sets of vertices and edges is:

$$G = (V, E) \tag{3.3}$$
$$V = \{A, B, C, D\} \tag{3.4}$$
$$E = \{(A, B), (B, A), (B, C), (C, B), (A, C), (C, A), (A, D), (D, A)\} \tag{3.5}$$

Graphs can also have loop. Loop are edges (A, A), $A \in V$, i.e. an edge connecting a vertex to itself. This is what graph 3.1 with a loop on vertex A:

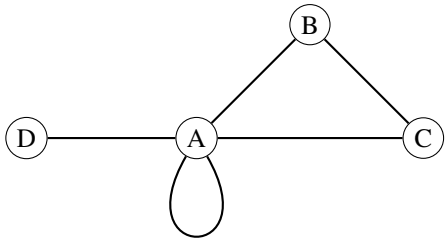


Figure 3.4: A graph with a loop in

3.2.1 Types of Graph

Connected Graphs

A connected graph is a graph where every vertex is connected to every other vertex. A directed graph is weakly connected if there is an edge between every vertex in the graph, and strongly connected if there is an edge in both directions between every vertex in the graph.

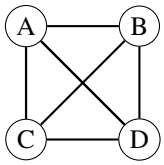


Figure 3.5: A connected graph

Simple Graphs

A graph is a simple graph if there exists at most one edge between each vertex, and there are no loop. This means that this is not a simple graph:

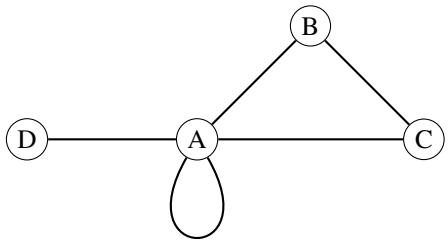


Figure 3.6: Not a simple graph, because it contains a loop

But this is:

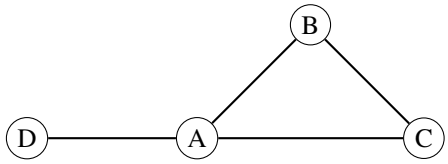


Figure 3.7: A simple graph

Complete Graphs

A complete graph is a graph with an edge between every pair of vertices. This means that, for any set of vertices, the complete graph using that set of vertices will create a graph isomorphic to all other complete graphs with that many vertices. The complete graph[2] with n vertices is written as K_n.

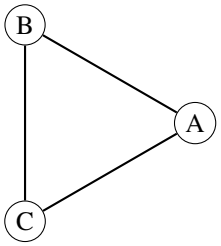


Figure 3.8: K_3

[2] A representative member of the equality class $[K_n]_{\cong}$

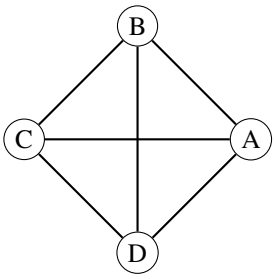


Figure 3.9: K_4

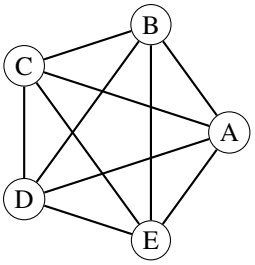


Figure 3.10: K_5

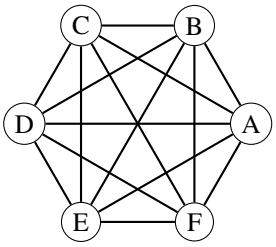


Figure 3.11: K_6

Theorem 3.2.1. *The complete graph K_n has $\frac{1}{2}\left(n^2 - n\right)$ edges.*

Proof. Each vertex in K_n is connected to every other vertex in K_n, therefore there must be n^2 combinations of vertices, (v_i, v_j). However, K_n doesn't have any loop, so (v_i, v_i) is not counted. For n vertices, the pairs (v_i, v_i) make up n edges, so there are $n^2 - n$ combination (v_i, v_j). However, K_n only has one edge between each pair of vertices, so (v_i, v_j) and (v_j, v_i) wouldn't both count, so the final number of vertex is $\frac{1}{2}\left(n^2 - n\right)$. □

3.2.2 Euler's Handshaking Lemma

Theorem 3.2.2 (Euler's Handshaking Lemma)**.** *The sum of directed graphs in a graph is equal to 2 times the number of edges.*

Proof. Each edge in a graph connects exactly 2 vertices, meaning that it adds one degree to each edge. This means that each edge adds two degree to the graph. □

3.2.3 Walks

A walk is a route along edges of a graph, but there are other parameters that can be put on a walk to make different types of walk.

A path is a walk where no vertex is more than once.

A trail is a walk where no edge is visited more than once.

A cycle is a walk where the first vertex visited is also the final vertex visited, and no other vertex is visited more than once.

A Hamiltonian cycle is a cycle where every vertex is visited.

Examples of Walks

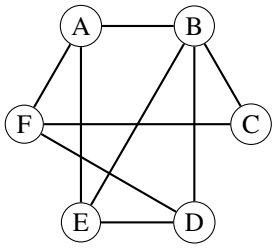


Figure 3.12: A more complicated graph

An example of a walk on graph 3.12 is A-B-A-F-C.

An example of a path on graph 3.12 is A-B-C-F.

An example of a trail on graph 3.12 is A-F-C-B.

An example of a cycle on graph 3.12 is A-B-C-F-A.

An example of a Hamiltonian cycle on graph 3.12 is E-A-F-C-B-D-E.

3.2.4 Subgraphs

A subgraph of $G = (V, E)$ is a graph $g = (v, e)$ where $v \subseteq V$ and $e \subseteq v^2 \subseteq E$. Since, for all sets S, $\varnothing \subseteq S$, the graph $(\varnothing, \varnothing)$ is a subgraph of all graphs. The graph $(\varnothing, \varnothing)$ is the set with no vertex and thus no edges. The statement 'g is a subgraph of G' is sometimes written as '$g \subseteq G$', or as '$g \sqsubseteq G$'.

Visually, a subgraph of a graph is the graph excluding any number of its vertices and edges, as long as the resulting graph is still a valid graph.

This is an example of a subgraph of graph 3.12:

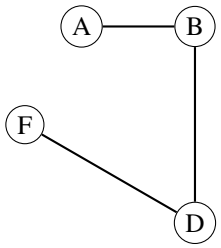


Figure 3.13: A subgraph of graph 3.12

Notice that, even though the edge A-F exists on graph 3.12, the edge does not have to exist in its subgraph.

To show more clearly that it is a subgraph of graph 3.12, this is it overlaid on top of graph 3.12:

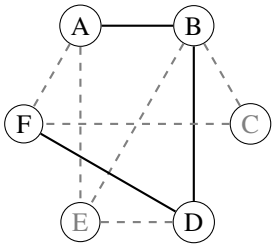


Figure 3.14: The subgraph (black) overlaid onto the original graph (grey)

3.2.5 Weighted Graphs

A weighted graph is a graph with weighted edges, i.e. an edge with a weight on. This weight could represent time taken to travel between two points, represented by vertices, for example.

This is an example of a weighted graph:

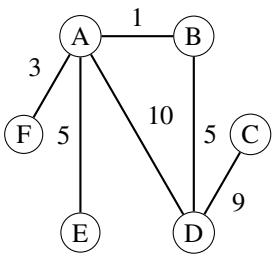


Figure 3.15: A weighted graph

Expressing a weighted graph as a set of vertices and edges ($G = (V, E)$) requires

turning each of the edges into an ordered triple, (v_1, v_2, w), where $v_1, v_2 \in V$, and w is the weight. w is usually a real number, but does not have to be.

3.2.6 Directed Graphs

A directed graph (AKA digraph) is a graph with directed edges. A directed edge is an edge that cannot be traversed in both directions. A one-way edge is drawn as an arrow in the direction that the edge can be traversed.

This is an example of a directed graph:

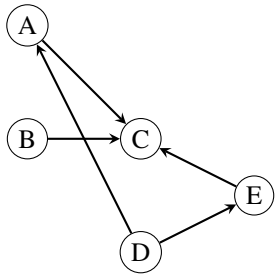


Figure 3.16: A directed graph

In the example above, B is connected to C, however C is not connected to B, since there is not a vertex C → B.

3.2.7 Graphs as Matrices

Adjacency Matrices

A graph can be represented as a matrix representing each edge. This is an example of a graph and its adjacency matrix:

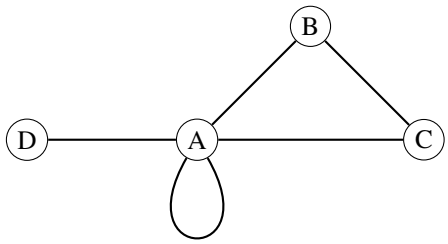


Figure 3.17: An example graph

	A	B	C	D
A	2	1	1	1
B	1	0	1	0
C	1	1	0	0
D	1	0	0	0

The adjacency matrix shows the number of edges between each apri of vertices. Because the graph is not directed, each edge X-Y also exists as Y-X, so the matrix is symmetric along its diagonal. The loop A-A is shown as 2 edges, because it is considered in both directions; the clockwise loop and anticlockwise loop count separately.

The A row/column has no 0s in it, meaning that there exists an edge between A and every other vertex. This also applies if there is only a 0 on the diagonal in that row/column.

The complete graph K_n has an adjacency matrix of only 1s, except for the diagonal, which is all 0s, meaning that every vertex is connected to every other, with no loop, which is the definition of a complete graph:

K_n	v_1	v_2	$\cdots$	v_n
v_1	0	1	$\cdots$	1
v_2	1	0	$\cdots$	1
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
v_n	1	1	$\cdots$	0

3.2.8 Trees

A tree is a connected graph with no cycles. This is an example of a tree:

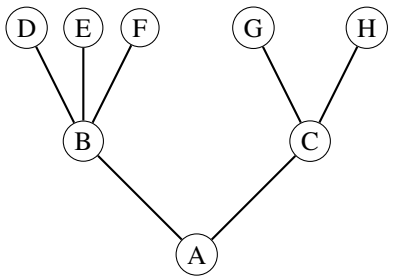


Figure 3.18: A tree

This condition implies that you can rearrange the graph into a graph with a root and branches connected to it. This means that this is not a tree (it is not connected):

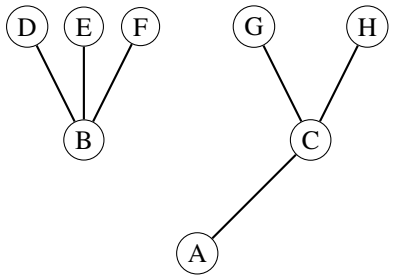


Figure 3.19: Not a tree, since there is no connection between A and B, D, E and F

This is also not a tree (it has a cycle, A-B-C-A):

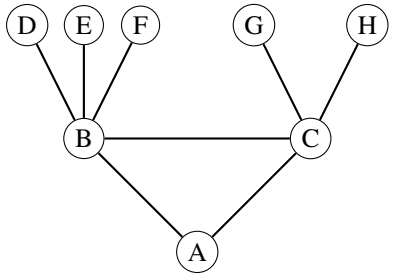


Figure 3.20: Not a tree, since there is a cycle, A-B-C-A

A spanning tree of a graph is a tree such that every vertex of the original graph is in the tree. For example, an example spanning tree of graph 3.12 (with the original graph in dashed grey) is:

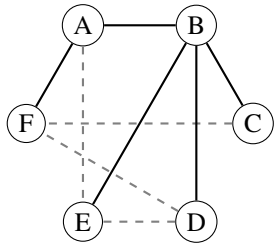


Figure 3.21: A spanning tree of graph 3.12

In a weighted graph, a minimum spanning tree (MST) is a spanning tree of a subgraph where the sum of all weights is the smallest possible value for a spanning tree of the graph.

3.3 Algorithms on Graphs

3.3.1 Kruskal's Algorithm

Description

Kruskal's algorithm is an algorithm for finding a minimum spanning tree of a weighted graph. The steps of the algorithm are:

1. Sort all of the vertices in ascending order of weight.
2. Add the first edge to the spanning tree.
3. Look at the next edge is the list, and add it to the spanning tree if and only if it does not create a cycle in the graph.
4. Repeat step 3. until all edges have either been added to the new spanning tree or rejected.

This is an example of Kruskal's algorithm on this graph:

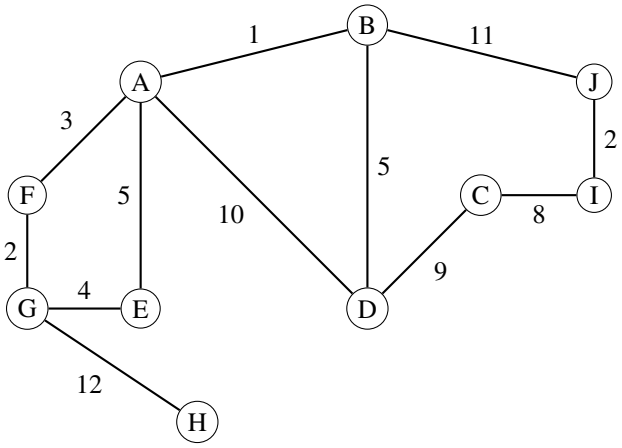


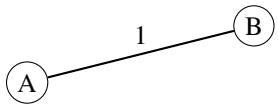
Figure 3.22: A weighted graph

The first step is to list all of the edges in ascending order of the weights:

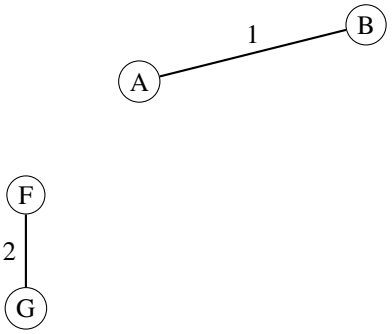
- A-B (1)
- F-G (2)
- I-J (2)
- A-F (3)
- E-G (4)
- A-E (5)

- B-D (5)
- C-I (8)
- C-D (9)
- A-D (10)
- B-J (11)
- G-H (12)

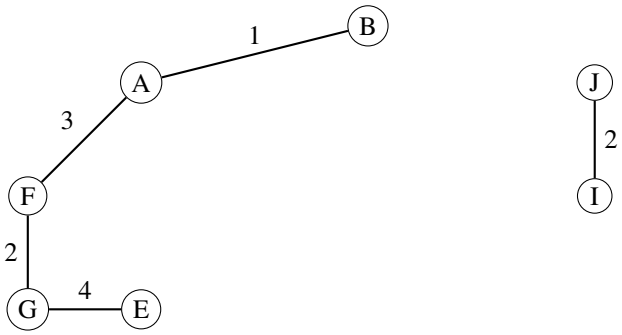
Then, we will add the first edge into a spanning tree:



Then, we will check to see if the second edge will create a cycle. The edge F-G will not create a cycle, so it can be added:

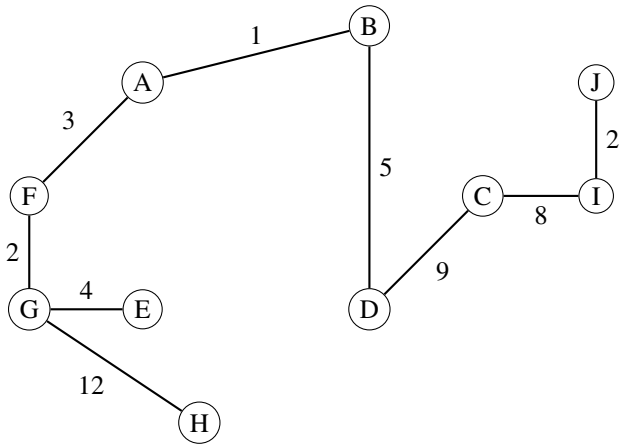


Continuing to edge A-E, we now have this spanning tree:



However, trying to add the edge A-E would create the cycle A-E-G-F-A, meaning A-E cannot be added while keeping the graph as a tree. This means that A-E will

be skipped, and we will continue to the next edge, B-D. Continuing the process, we eventually get the spanning tree:



In this graph, there are no cycles, it is fully connected, and contains every vertex of the original graph, meaning that it is a spanning tree of the original graph.

Theorem 3.3.1. *Applying Kruskal's algorithm to a weighted connected graph will create a spanning tree.*

Proof. By the definition of a tree, a tree must be a connected graph and must not have any cycles.

By the definition of a spanning tree, the tree must contain every vertex of the original graph.

Thus, to prove the initial statement, we will prove each of the parts separately:

- **The graph is connected** - Kruskal's algorithm will only reject an edge if that edge forms a cycle. If the edge would form a cycle, that means that there is already a path[3] around where the edge would have connected. This is because, for an edge V_1-V_2, if V_1-V_2 joins up a cycle, then there exists a path V_1-$\cdots$-V_2; the opposite way around the circle formed by the cycle. For a directed graph, the proof still applies, but it may be weakly connected.
- **The graph does not contain any cycles** - step 2. of Kruskal's algorithm (see above) states that each edge should be added *"if and only if it does not create a cycle in the graph"*, thus no edge can create a cycle.
- **The graph contains every vertex of the original graph** - the original graph is stated in the initial statement to be a connected graph, so every edge connects two vertices in the graph. The only edges not added are those that create a cycle, thus, using the same logic as in the proof of the graph not containing any cycles, there must exist another path around the cycle, meaning that it must be connected.

[3]Here, we count directed edges as working both ways. If the directed edges do not permit an actual path, then the graph is weakly connected. This only applies to directed graphs.

Thus, since all of the conditions for the resulting graph being a spanning tree hold, the initial statement is true. □

It can also be proved that this spanning tree is a minimum spanning tree, but we will not prove that here.

Implementation

```
// Takes a graph and uses Kruskal's algorithm to find a
// minimum spanning tree of the graph.
// Example:
//     Input:  GRAPH([A, B, C, D],
//                   [(A, B, 2),
//                    (B, C, 3),
//                    (C, D, 1),
//                    (A, D, 2),
//                    (A, C, 4)])
//     Output: GRAPH([A, B, C, D]
//                   [(A, B, 2),
//                    (B, C, 3),
//                    (C, D, 1)])
FUNC kruskals_algorithm(graph):
    sortedEdges = sort(graph.edges)

    newGraph = NEW GRAPH([], [])

    FOR EACH edge IN graph.edges:
        checkGraph = graph
        checkGraph.edges.PUSH(edge)

        IF (NOT checkGraph.HAS_CYCLE):  // Use DFS to find cycles
            newGraph.edges.PUSH(edge)
        END IF
    END FOR

    RETURN newGraph
END FUNC
```

3.3.2 Primm's Algorithm

3.4 Route Inspection

3.5 The Travelling Salesman Problem

3.6 Linear Programming

3.7 The Simplex Algorithm

3.8 Critical Path Analysis

Chapter 4

Decision Year II

Decision II Textbook (D2)

4.1 Transportation Problems

4.2 Allocation Problems

4.3 Flows in Networks I

4.4 Flows in Networks II

4.5 Dynamic Programming

4.6 Game Theory

4.7 Recurrence Relations

4.8 Decision Analysis

Chapter 5

Further Mechanics Year I

Further Mechanics I Textbook (FM1)

5.1 Momentum and Impulse

5.1.1 Momentum

Momentum is the property of an object given by its mass and velocity; the more massive an object is and the faster it is moving, the more momentum it has. A car moving slowly won't have much momentum since its velocity is low, but if it is moving quickly it will. A bullet thrown by hand wouldn't have much momentum, since it is very small and not very fast moving, but when shot, it would have a lot of momentum, since it is moving very fast, despite being small.

Since momentum is proportional to velocity and momentum, the equation is given as:

$$\mathbf{p} = m\mathbf{v} \tag{5.1}$$

Where $\mathbf{p}$ is momentum, m is the mass of the object, and $\mathbf{v}$ is the velocity. See equation B.43.

Units

In SI units, mass is in kilograms (kg), and momentum is in metres per second (ms^{-1}). Since the two values are multiplied, this means that the units of momentum must be kilogram metres per second (kgms^{-1}).

However, since newtons are defined as kilogram metres per second squared (kgms^{-2}), newton seconds (Ns) would be kilogram metres per second (kgms^{-1}), which is the unit for momentum, so momentum can either be written in kgms^{-1} or in Ns.

Force/Time Definition

Since the units of momentum can be written as newton seconds (the unit of force $\times$ the unit of time), it looks like it might be possible to write momentum as a product of force

and time.

Theorem 5.1.1. *Momentum is the product of force and time:*

$$\mathbf{p} = \mathbf{F}t \tag{5.2}$$

Proof. We start the proof with the following identities:

$$\mathbf{F} = m\mathbf{a} \tag{5.3}$$

$$\mathbf{a} = \frac{\mathbf{v}}{t} \tag{5.4}$$

$$\mathbf{p} = m\mathbf{v} \tag{5.5}$$

From this, $\mathbf{F}$ can be written in terms of m, $\mathbf{v}$, and t by substituting equation 5.4 into equation 5.3 we get:

$$\mathbf{F} = m\frac{\mathbf{v}}{t}$$

Multiplying both sides by t gives:

$$\mathbf{F}t = m\mathbf{v}$$

And finally, the $m\mathbf{v}$ can be replaced with the identity in equation 5.5 to get the result:

$$\mathbf{p} = \mathbf{F}t$$

□

This means that the momentum given to a particle after exerting a force $\mathbf{F}$ for t time will be $\mathbf{F}t$.

5.1.2 Impulse

Impulse ($\mathbf{I}$) is the change of momentum between two moments, i.e.:

$$\mathbf{I} = \Delta\mathbf{p} = m(\mathbf{v} - \mathbf{u}) \tag{5.6}$$

Where $\mathbf{v}$ is final momentum, and $\mathbf{u}$ is the initial momentum.

5.1.3 Momentum and Impulse as Vectors

5.2 Work, Energy and Power

5.3 Elastic Strings and Springs

5.4 Elastic Collisions in One Dimension

5.5 Elastic Collisions in Two Dimensions

Chapter 6

Further Mechanics Year II

Further Mechanics Textbook (FM2)

6.1 Circular Motion

6.2 Centre of Mass of Plane Figures

6.3 Further Centres of Mass

6.4 Kinematics

6.5 Dynamics

Chapter 7

Further Pure Mathematics Year I

Further Pure I Textbook (FP1)

7.1 Vectors

7.1.1 The Vector Product

For two 3D vectors, $\mathbf{a}$ and $\mathbf{b}$, the vector product (also called the cross product) is written $\mathbf{a} \times \mathbf{b}$. The vector product is defined as:

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin\theta \, \hat{\mathbf{n}} \tag{7.1}$$

Where θ is the angle between the two angles (this can be found using the dot product), and $\hat{\mathbf{n}}$ is the unit vector perpendicular to $\mathbf{a}$ and $\mathbf{b}$. However, since $\mathbf{a}$ and $\mathbf{b}$ are both 3D, there are two opposite unit vectors that are both perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ ($\hat{\mathbf{n}}$ and $-\hat{\mathbf{n}}$).

The 'right-hand rule' can be used to find the direction of $\hat{\mathbf{n}}$. Imagining your index finger as $\mathbf{a}$, your middle finger as $\mathbf{b}$, the thumb of you right hand will point in the direction of $\hat{\mathbf{n}}$. This means that swapping $\mathbf{a}$ and $\mathbf{b}$ will flip the direction of $\hat{\mathbf{n}}$. Therefore, $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$. This property is called anticommutatitivity (opposed to commutatitivity, such as for addition, where $a + b = b + a$).

Another way of thinking about flipping the direction of the vector product is by changing the direction of the angle measurement. θ is measured as the smallest angle between $\mathbf{a}$ and $\mathbf{b}$, meaning that $\sin\theta > 0$. However, flipping $\mathbf{a}$ and $\mathbf{b}$ would be the same as measuring the angle in the other direction, which would give an angle $2\pi - \theta$ (in radians). Using the identity:

$$\sin\theta = -\sin(2\pi - \theta) \tag{7.2}$$

Measuring the angle from the opposite direction would give a negative answer, which is the same as it would when negating $\hat{\mathbf{n}}$.

Vector Product Indentities

From definition 7.1, the following identities can be found for the product of the basis vectors **i**, **j**, and **k**:

$$\mathbf{i}\times\mathbf{i}=\mathbf{j}\times\mathbf{j}=\mathbf{k}\times\mathbf{k}=0 \tag{7.3}$$

$$\mathbf{i}\times\mathbf{j}=-(\mathbf{j}\times\mathbf{i})=\mathbf{k} \tag{7.4}$$

$$\mathbf{j}\times\mathbf{k}=-(\mathbf{k}\times\mathbf{j})=\mathbf{i} \tag{7.5}$$

$$\mathbf{k}\times\mathbf{i}=-(\mathbf{i}\times\mathbf{k})=\mathbf{j} \tag{7.6}$$

Theorem 7.1.1. *Using any two non-zero vectors,* **a** *and* **b**, $\mathbf{a}\times\mathbf{b}=0$ *if and only if they are parallel.*

Proof. Given definition 7.1, $|\mathbf{a}\times\mathbf{b}|$ can be found as the product of the magnitudes of each of of the parts of the definition:

$$|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}||\sin\theta||\hat{\mathbf{n}}| \tag{7.7}$$

Given that $0\leq\theta\leq\pi$, $\sin\theta\geq 0$, so $|\sin\theta|=\sin\theta$. We also know that $|\hat{\mathbf{n}}|$ is a unit vector by definition, so $|\hat{\mathbf{n}}|=1$, so the formula for the magnitude of $|\mathbf{a}\times\mathbf{b}|$ simplifies to:

$$|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta \tag{7.8}$$

Thus, (since **a** and **b** are non-zero), $|\mathbf{a}\times\mathbf{b}|=0$ if and only if $\sin\theta=0$, which would mean that $\theta=0$ or $\theta=\pi$. If $\theta=0$ or $\theta=\pi$, then **a** and **b** must be parallel, since $\theta=0$ means that $\sin\theta=0$. □

Theorem 7.1.2. *Using any two vectors,* **a** *and* **b**, $\mathbf{a}\times\mathbf{b}=|\mathbf{a}||\mathbf{b}|$ *if and only if they are perpendicular.*

Proof. Given definition 7.1, $|\mathbf{a}\times\mathbf{b}|$ can be found as the product of the magnitudes of each of of the parts of the definition:

$$|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}||\sin\theta||\hat{\mathbf{n}}| \tag{7.9}$$

Given that $0\leq\theta\leq\pi$, $\sin\theta\geq 0$, so $|\sin\theta|=\sin\theta$. We also know that $|\hat{\mathbf{n}}|$ is a unit vector by definition, so $|\hat{\mathbf{n}}|=1$, so the formula for the magnitude of $|\mathbf{a}\times\mathbf{b}|$ simplifies to:

$$|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta \tag{7.10}$$

Thus, $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$ if and only if $\sin\theta=1$, which would mean that $\theta=\frac{\pi}{2}$, which means that they are perpendicular, since $\frac{\pi}{2}$ is a right-angle. □

Theorem 7.1.3. *Where the vectors* **a** *and* **b** *are defined as follows:*

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}$$

$$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Their vector product, $\mathbf{a} \times \mathbf{b}$*, is:*

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \tag{7.11}$$

Or in column form:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} \tag{7.12}$$

Proof. We'll begin with expanding $\mathbf{a} \times \mathbf{b}$ using their $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ forms:

$$\mathbf{a} \times \mathbf{b} = (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \times (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) \tag{7.13}$$

From there, the two trinomials can be expanded normally:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} = {} & a_1b_1(\mathbf{i} \times \mathbf{i}) + a_1b_2(\mathbf{i} \times \mathbf{j}) + a_1b_3(\mathbf{i} \times \mathbf{k}) \\ & + a_2b_1(\mathbf{j} \times \mathbf{i}) + a_2b_2(\mathbf{j} \times \mathbf{j}) + a_2b_3(\mathbf{j} \times \mathbf{k}) \\ & + a_3b_1(\mathbf{k} \times \mathbf{i}) + a_3b_2(\mathbf{k} \times \mathbf{j}) + a_3b_3(\mathbf{k} \times \mathbf{k}) \end{aligned} \tag{7.14}$$

From here, we can use the identities 7.3, 7.4, 7.5, 7.6 to simplify each term of the expanded trinomial:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} = {} & a_1b_2(\mathbf{k}) + a_1b_3(-\mathbf{j}) \\ & + a_2b_1(-\mathbf{k}) + a_2b_3(\mathbf{i}) \\ & + a_3b_1(-\mathbf{i}) + a_3b_2(\mathbf{j}) \\ = {} & (a_2b_3 - a_3b_1)\mathbf{i} + (a_3b_2 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \end{aligned} \tag{7.15}$$

Which is the form in equation 7.11. □

This can also be written as the determinant of a 3×3 matrix:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \mathbf{i}\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} + \mathbf{j}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \mathbf{k}\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \tag{7.16}$$

7.1.2 Finding Areas of Triangles and Parallelograms with Vectors

The area of a triangle with legs a and b and an angle θ between them is:

$$A = \frac{1}{2}ab\sin\theta \tag{7.17}$$

This closely resembles the formula for the vector product, where $a = |\mathbf{a}|$ and $b = |\mathbf{b}|$. The using those definitions of a and b, the formula for the area of a triangle is:

$$\begin{aligned} A &= \frac{1}{2}|\mathbf{a}||\mathbf{b}|\sin\theta \\ &= \frac{1}{2}|\mathbf{a} \times \mathbf{b}| \end{aligned} \tag{7.18}$$

Thus, this formula can be used to derive the area of a triangle from its side, where its sides are given by vectors. This formula can also be extended to parallelograms; the area of a parallelogram with sides a and b is twice the area of a triangle with sides a and b, so its area would be given as:

$$A = |\mathbf{a} \times \mathbf{b}| \tag{7.19}$$

7.1.3 Example Questions

Question 7.1.1. *From FP1; question 1A/1f*: Simplify $(3\mathbf{i}+\mathbf{j}-\mathbf{k})\times 2\mathbf{j}$.

Answer 7.1.1: $(3\mathbf{i}+\mathbf{j}-\mathbf{k})\times 2\mathbf{j}$ can be written as a 3×3 matrix determinant:

$$(3\mathbf{i}+\mathbf{j}-\mathbf{k})\times 2\mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -1 \\ 0 & 2 & 0 \end{vmatrix}$$

Which can then be separated into:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -1 \\ 0 & 2 & 0 \end{vmatrix} = \mathbf{i}\begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} + \mathbf{j}\begin{vmatrix} 3 & -1 \\ 0 & 0 \end{vmatrix} + \mathbf{k}\begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix}$$

Then, each of the matrix determinants can be found to fully simplify the vector product:

$$\begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} = 1\times 0 - (-1)\times 2 = 2$$

$$\begin{vmatrix} 3 & -1 \\ 0 & 0 \end{vmatrix} = 3\times 0 - (-1)\times 0 = 0$$

$$\begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix} = 3\times 2 - 1\times 0 = 6$$

So the final vector is $2\mathbf{i}+6\mathbf{k} = \boxed{2\mathbf{i}+6\mathbf{k}}$.

Question 7.1.2. *From FP1; question 1A/9*: Given that $\mathbf{a} = -\mathbf{i}+2\mathbf{j}-5\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i}-2\mathbf{j}+\mathbf{k}$, find:

a) $\mathbf{a}\cdot\mathbf{b}$

b) $\mathbf{a}\times\mathbf{b}$

c) The unit vector in the direction $\mathbf{a}\times\mathbf{b}$

Answer 7.1.2:

a)

$$\begin{aligned} \mathbf{a}\cdot\mathbf{b} &= (-\mathbf{i}+2\mathbf{j}-5\mathbf{k})\cdot(5\mathbf{i}-2\mathbf{j}+\mathbf{k}) \\ &= (-1\times 5)+(2\times -2)+(-5\times 1) \\ &= -5-4-5 \\ &= \boxed{-14} \end{aligned}$$

b)

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ -1 & 2 & -5\\ 5 & -2 & 1\end{vmatrix}\\ &= \mathbf{i}\begin{vmatrix}2 & -5\\ -2 & 1\end{vmatrix} + \mathbf{j}\begin{vmatrix}-1 & -5\\ 5 & 1\end{vmatrix} + \mathbf{k}\begin{vmatrix}-1 & 2\\ 5 & -2\end{vmatrix}\\ &= \boxed{-8\mathbf{i} - 24\mathbf{j} - 8\mathbf{k}}\end{aligned}$$

c) The unit vector in direction $\mathbf{v}$ is $\frac{\mathbf{v}}{|\mathbf{v}|}$. Using the result from part b:

$$\begin{aligned}|\mathbf{a}\times\mathbf{b}| &= \sqrt{(-8)^2 + (-24)^2 + (-8)^2}\\ &= 8\sqrt{11}\end{aligned}$$

Therefore, the unit vector in direction $\mathbf{a}\times\mathbf{b}$ is:

$$\frac{1}{8\sqrt{11}}(-8\mathbf{i} - 24\mathbf{j} - 8\mathbf{k}) = \boxed{-\frac{1}{\sqrt{11}}(\mathbf{i} + 3\mathbf{j} + \mathbf{k})}$$

Question 7.1.3. *From FP1; question 1A/15*: If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$, show that:

$$\mathbf{a}\times\mathbf{b} = \mathbf{b}\times\mathbf{c} = \mathbf{c}\times\mathbf{a} \tag{7.20}$$

Answer 7.1.3: We will begin by multiplying $\mathbf{a}$ by $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$:

$$\begin{aligned}\mathbf{a}\times(\mathbf{a}+\mathbf{b}+\mathbf{c}) &= \mathbf{a}\times 0\\ \mathbf{a}\times\mathbf{a} + \mathbf{a}\times\mathbf{b} + \mathbf{a}\times\mathbf{c} &= 0\end{aligned}$$

Since for any vector $\mathbf{v}$, $\mathbf{v}\times\mathbf{v} = 0$, the $\mathbf{a}\times\mathbf{a}$ can be removed. Then, using the anticommutatitivity of the vector product ($\mathbf{v}\times\mathbf{u} = -(\mathbf{u}\times\mathbf{v})$), the $\mathbf{b}\times\mathbf{c}$ can be negated then moved to the right hand side:

$$\begin{aligned}\mathbf{a}\times\mathbf{b} + \mathbf{a}\times\mathbf{c} &= 0\\ \mathbf{a}\times\mathbf{b} - \mathbf{c}\times\mathbf{a} &= 0\\ \mathbf{a}\times\mathbf{b} &= \mathbf{c}\times\mathbf{a}\end{aligned}$$

Now we will repeat the process by multiplying $\mathbf{b}$ by $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$:

$$\begin{aligned}\mathbf{b}\times(\mathbf{a}+\mathbf{b}+\mathbf{c}) &= \mathbf{a}\times 0\\ \mathbf{b}\times\mathbf{a} + \mathbf{b}\times\mathbf{b} + \mathbf{b}\times\mathbf{c} &= 0\\ \mathbf{b}\times\mathbf{a} + \mathbf{b}\times\mathbf{c} &= 0\\ \mathbf{b}\times\mathbf{a} - \mathbf{c}\times\mathbf{b} &= 0\\ \mathbf{b}\times\mathbf{a} &= \mathbf{c}\times\mathbf{b}\\ \mathbf{a}\times\mathbf{b} &= \mathbf{b}\times\mathbf{c}\end{aligned}$$

Thus, since $\mathbf{a} \times \mathbf{b} = \mathbf{c} \times \mathbf{a}$ and $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}$:

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \tag{7.21}$$

□

7.2 Conic Sections I

7.3 Conic Sections II

7.4 Inequalities

7.5 The *t*-Formulae

7.6 Taylor Series

7.7 Methods in Calculus

7.8 Numerical Methods

7.9 Reducible Differential Equations

Chapter 8

Further Pure Mathematics Year II

Further Pure II Textbook (FP2)

8.1 Number Theory

8.2 Groups

8.3 Complex Numbers

8.4 Recurrence Relations

8.5 Matrix Algebra

8.6 Integration Techniques

Chapter 9

Further Statistics Year I

Further Statistics I Textbook (FS1)

9.1 Discrete Random Variables

9.2 Poisson Distributions

9.3 Geometric and Negative Binomial Distributions

9.4 Hypothesis Testing

9.5 Central Limit Theorem

9.6 χ^2 Tests

9.7 Probability Generating Functions

9.8 Quality of Tests

Chapter 10

Further Statistics Year II

Further Statistics II Textbook (FS2)

10.1 Linear Regression

10.2 Correlation

10.3 Continuous Distribution

10.4 Combinations of Random Variables

10.5 Estimation, Confidence Intervals and Tests Using a Normal Distribution

10.6 Further Hypothesis Tests

10.7 Confidence Intervals and Tests Using the t-Distribution

Appendix A

Appendix: Formula Sheet

This formula sheet contains the formulae from the Pearson Edexcel formula sheet. See original here.

A.1 Pure Mathematics

A.1.1 Summations

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1) \tag{A.1}$$

$$\sum_{r=1}^{n} r^3 = \frac{1}{6}n^2(n+1)^2 \tag{A.2}$$

A.1.2 Matrix Transformations

Anticlockwise rotation through θ about O:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{A.3}$$

Reflection in the line $y = (\tan\theta)x$:

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \tag{A.4}$$

A.1.3 Area of a Sector

$$A = \frac{1}{2}\int r^2 d\theta \quad \text{(polar coordinates)} \tag{A.5}$$

A.1.4 Complex Numbers

$$[r\cos\theta + i\sin\theta]^n = r^n(\cos n\theta + i\sin n\theta) \tag{A.6}$$

The roots of $z^n = 1$ are given by

$$z = e^{\frac{2\pi ki}{n}}, \text{ for } k = 0, 1, 2, \ldots, n-1 \tag{A.7}$$

A.1.5 Maclaurin's and Taylor's Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^r}{r!}f^{(r)}(0) + \cdots \tag{A.8}$$

$$e^x = \exp(x) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^r}{r!} + \cdots \quad \text{for all } x \tag{A.9}$$

$$\ln(n+1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^r\frac{x^r}{r} + \cdots \quad (-1 < x \leq 1) \tag{A.10}$$

$$\cos x = 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^{r+1}\frac{x^{2r+1}}{(2r+1)!} + \cdots \quad \text{for all } x \tag{A.11}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^r\frac{x^{2r}}{(2r)!} \quad \text{for all } x \tag{A.12}$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^r\frac{x^{2r+1}}{2r+1} + \cdots \quad (-1 \leq x \leq 1) \tag{A.13}$$

A.1.6 Vectors

The resolved part of $\mathbf{a}$ in the direction of $\mathbf{b}$ is

$$\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}} \tag{A.14}$$

Vector Product

$$\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\hat{\mathbf{n}} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix} = \begin{bmatrix}a_2b_3 - a_3b_2\\ a_3b_1 - a_1b_3\\ a_1b_2 - a_2b_1\end{bmatrix} \tag{A.15}$$

$$\mathbf{a}\cdot(\mathbf{b}b\times\mathbf{c}) = \begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix} = \mathbf{b}(\mathbf{c}\times\mathbf{a}) = \mathbf{c} = (\mathbf{a}\times\mathbf{b}) \tag{A.16}$$

If A is the point with position vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and the direction vector $\mathbf{b}$ is given by $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, then the straight line through A with direction vector $\mathbf{b}$ has Cartesian equation:

$$\frac{x-a_1}{b_1} = \frac{y-a_2}{b_2} = \frac{z-a_3}{b_3} = \lambda \tag{A.17}$$

The plane through A with normal vector $\mathbf{n} = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$ has Cartesian equation:

$$n_1x + n_2y + n_3z + d = 0 \text{ where } d = -\mathbf{a}\cdot\mathbf{n} \tag{A.18}$$

The plane through non-collinear points A, B and C has vector equation:

$$\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b}-\mathbf{a}) + \mu(\mathbf{c}-\mathbf{a}) = (1-\lambda-\mu)\mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c} \tag{A.19}$$

The plane through A and parallel to $\mathbf{b}$ and $\mathbf{c}$ as vector equation:

$$\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c} \tag{A.20}$$

The perpendicular distance of (α, β, γ) from $n_1x + n_2y + n_3z + d = 0$ is:

$$\frac{|n_1\alpha + n_2\beta + n_3\gamma + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} \tag{A.21}$$

A.1.7 Hyperbolic Functions

$$\cosh^2 x - \sinh^2 x = 1 \tag{A.22}$$

$$\sinh 2x = 2\sinh x \cosh x \tag{A.23}$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x \tag{A.24}$$

$$\cosh^{-1} = \ln\left\{x + \sqrt{x^2-1}\right\} \quad (x \geq 1) \tag{A.25}$$

$$\sinh^{-1} = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1) \tag{A.26}$$

A.1.8 Conics

Ellipses

Standard Form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{A.27}$$

Parametric Form:

$$(a\cos\theta, b\sin\theta) \tag{A.28}$$

Eccentricity

$$\begin{gathered} e < 1 \\ b^2 = a^2(1-e^2) \end{gathered} \tag{A.29}$$

Foci:

$$(\pm ae, 0) \tag{A.30}$$

Directrices:

$$x = \pm\frac{a}{e} \tag{A.31}$$

Asymptotes: none

Parabolas

Standard Form:

$$y = 4ax \tag{A.32}$$

Parametric Form:

$$(at^2, 2at) \tag{A.33}$$

Eccentricity:

$$e = 1 \tag{A.34}$$

Foci:

$$(a, 0) \tag{A.35}$$

Directrices:

$$x = -a \tag{A.36}$$

Asymptotes: none

Hyperbolae

Standard Form:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{A.37}$$

Parametric Forms:

$$(a \sec\theta, b \tan\theta) \tag{A.38}$$

$$(\pm a \cosh\theta, b \sinh\theta) \tag{A.39}$$

Eccentricity:

$$\begin{gathered} e > 1 \\ b^2 = a^2(e^2 - 1) \end{gathered} \tag{A.40}$$

Foci:

$$(\pm ae, 0) \tag{A.41}$$

Directrices:

$$x = \pm\frac{a}{e} \tag{A.42}$$

Asymptotes:

$$\frac{x}{a} = \pm\frac{y}{b} \tag{A.43}$$

Rectangular Hyperbolae

Standard Form:

$$xy = c^2 \tag{A.44}$$

Parametric Form:

$$\left(ct, \frac{c}{t}\right) \tag{A.45}$$

Eccentricity:

$$e = \sqrt{2} \tag{A.46}$$

Foci:

$$(\pm\sqrt{2}c, \pm\sqrt{2}c) \tag{A.47}$$

Directrices:

$$x + y = \pm\sqrt{2}c \tag{A.48}$$

Asymptotes:

$$x = 0,\ y = 0 \tag{A.49}$$

A.1.9 Differentiation

$f(x)$	$f'(x)$	
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	(A.50)
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	(A.51)
$\arctan x$	$\frac{1}{\sqrt{1+x^2}}$	(A.52)
$\sinh x$	$\cosh x$	(A.53)
$\cosh x$	$\sinh x$	(A.54)
$\tanh x$	$\operatorname{sech}^2 x$	(A.55)
$\operatorname{arsinh} x$	$\frac{1}{1+x^2}$	(A.56)
$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x^2-1}}$	(A.57)
$\operatorname{artanh} x$	$\frac{1}{1-x^2}$	(A.58)
		(A.59)

A.1.10 Integration (+ constant; $a > 0$. where relevant)

$f(x)$	$\int f(x)dx$	
$\sinh x$	$\cosh x$	(A.60)
$\cosh x$	$\sinh x$	(A.61)
$\tanh x$	$\ln \cosh x$	(A.62)
$\frac{1}{\sqrt{a^2 - x^2}}$	$\sin^- 1\left(\frac{x}{a}\right) + c \quad (\lvert x\rvert < a)$	(A.63)
$\frac{1}{a^2 + x^2}$	$\frac{1}{a}\tan^- 1\left(\frac{x}{a}\right) + c$	(A.64)
$\frac{1}{x^2 - a^2}$	$\cosh^- 1(\frac{x}{a})$ or $\ln\left\{x + \sqrt{x^2 - a^2} + c \quad (x > a)\right\}$	(A.65)
$\frac{1}{\sqrt{a^2 + x^2}}$	$\sinh^{-1}\left(\frac{x}{a}\right)$ or $\ln\left\{x + \sqrt{x^2 + a^2}\right\} + c$	(A.66)
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + c \quad (\lvert x\rvert < a)$	(A.67)
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a-x}{a+x}\right\rvert + c$	(A.68)

A.1.11 Arc Length

$$s = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \quad \text{(Cartesian coordinates)} \tag{A.69}$$

$$s = \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \quad \text{(parametric form)} \tag{A.70}$$

$$s = \int \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad \text{(polar form)} \tag{A.71}$$

A.1.12 Surface Area of Revolution

$$s_x = 2\pi \int y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \quad \text{(Cartesian coordinates)} \tag{A.72}$$

$$s_x = 2\pi \int y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \quad \text{(parametric form)} \tag{A.73}$$

$$s_x = 2\pi \int r\sin\theta\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad \text{(polar form)} \tag{A.74}$$

A.2 Mechanics

A.2.1 Constant Acceleration

$$s = ut + \frac{1}{2}at^2 \qquad \mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 \tag{A.75}$$

$$s = vt - \frac{1}{2}at^2 \qquad \mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2 \tag{A.76}$$

$$v = u + at \qquad \mathbf{v} = \mathbf{u} + \mathbf{a}t \tag{A.77}$$

$$s = \frac{1}{2}(u + v)t \qquad \mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t \tag{A.78}$$

$$v^2 = u^2 + 2as \tag{A.79}$$

A.2.2 Centres of Mass

For uniform bodies:

Triangular lamina: $\frac{2}{3}$ along median from vertex (A.80)

Solid hemisphere, radius r: $\frac{3}{8}r$ from centre (A.81)

Hemispherical shell: $\frac{2}{3}$ along median from vertex (A.82)

Circular arc, radius r angle at centre 2α: $\frac{r\sin\alpha}{\alpha}$ from centre (A.83)

Sector of circle, radius r angle at centre 2α: $\frac{2r\sin\alpha}{3\alpha}$ from centre (A.84)

Solid cone or pyramid of height h: $\frac{1}{4}h$ above the base on the line from the centre of the base to the vertex (A.85)

Conical shell of height h: $\frac{1}{3}h$ above the base on the line from the centre of the base to the vertex (A.86)

A.3 Probability and Statistics

A.3.1 Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{A.87}$$

$$P(A \cap B) = P(A) \times P(B|A) \tag{A.88}$$

A.3.2 Standard Deviation

$$\sqrt{\frac{\sum(x - \bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \tag{A.89}$$

A.3.3 Discrete Distributions

Binomial $\mathrm{B}(n, p)$

$P(X = x)$:

$$\binom{n}{x} p^x (1-p)^{n-x} \tag{A.90}$$

Mean:

$$np \tag{A.91}$$

Variance:

$$np(1-p) \tag{A.92}$$

Poisson $\mathrm{Po}(\lambda)$

$P(X = x)$:

$$e^{-\lambda} \frac{\lambda^x}{x!} \tag{A.93}$$

Mean:

$$\lambda \tag{A.94}$$

Variance:

$$\lambda \tag{A.95}$$

Sampling Distributions

For a random sample $X_1, X_2, \ldots, X_n$ of n independent observations from a distribution having mean μ and a variance σ^2:

$$\bar{X} \text{ is an unbiased estimator of } \mu, \text{ with } \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} \tag{A.96}$$

$$S^2 \text{ is an unbiased estimator of } \sigma^2, \text{ where } S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1} \tag{A.97}$$

For a random sample of n observations from $\mathrm{N}(\mu, \sigma^2)$:

$$\frac{\bar{X} - \mu}{\sigma} \sim \mathrm{N}(0, 1) \tag{A.98}$$

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1} \tag{A.99}$$

A.3.4 Distribution-Free (Non-Parametric) Tests

Contingency tables:

$$\sum \frac{(O_i - E_i)^2}{E_i} \text{ is approximately distributed as } \chi^2 \tag{A.100}$$

Appendix B

Apendix: Non-Formula Sheet

B.1 Pure Mathematics

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{B.1}$$

B.1.1 Vieta's Formulae

For a quadratic $ax^2 + bx + c$ with roots α, β:

$$\sum \alpha = \alpha + \beta = -\frac{b}{a} \tag{B.2}$$

$$\alpha\beta = \frac{c}{a} \tag{B.3}$$

For a cubic $ax^3 + bx^2 + cx + d$ with roots α, β, γ:

$$\sum \alpha = \alpha + \beta + \gamma = -\frac{b}{a} \tag{B.4}$$

$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \tag{B.5}$$

$$\alpha\beta\gamma = -\frac{d}{a} \tag{B.6}$$

For a quartic $ax^4 + bx^3 + cx^2 + dx + e$ with roots α, β, γ, δ:

$$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{b}{a} \tag{B.7}$$

$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a} \tag{B.8}$$

$$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a} \tag{B.9}$$

$$\alpha\beta\gamma\delta = \frac{e}{a} \tag{B.10}$$

B.1.2 Integration

$$\int u\frac{du}{dx}dx = uv - \int v\frac{du}{dx}dx \tag{B.11}$$

$$\int \frac{f'(x)}{f(x)} = \ln|f(x)| + c \tag{B.12}$$

$\tan x$	$\ln\lvert\sec x\rvert + c$	(B.13)
$\cot x$	$\ln\lvert\sin x\rvert + c$	(B.14)
$\operatorname{cosec} x$	$-\ln\lvert\operatorname{cosec} x + \cot x\rvert = \ln\left\lvert\tan\frac{1}{2}x\right\rvert + c$	(B.15)
$\sec x$	$\ln\lvert\sec x + \tan x\rvert = \ln\left\lvert\tan\left(\frac{1}{2}x + \frac{1}{4}\pi\right)\right\rvert + c$	(B.16)
$\sec^2 x$	$\tan x + c$	(B.17)
$\tanh x$	$\ln\cosh x + c$	(B.18)

B.1.3 Binomial Series

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + b^n \quad (n \in \mathbb{N}) \tag{B.19}$$

Arithmetic Series

$$S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n[2a + (n-1)d] \tag{B.20}$$

Geometric Series

$$S_n = \frac{a(1-r^n)}{1-r} \tag{B.21}$$

$$S_\infty = \frac{a}{1-r} \tag{B.22}$$

B.1.4 Trigonometry: Small Angles

$$\sin\theta \approx \theta \tag{B.23}$$

$$\cos\theta \approx 1 - \frac{\theta^2}{2} \tag{B.24}$$

$$\tan\theta \approx \theta \tag{B.25}$$

B.1.5 Trigonometric Identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \tag{B.26}$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B \tag{B.27}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} \tag{B.28}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2} \tag{B.29}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} \tag{B.30}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2} \tag{B.31}$$

B.1.6 Differentiation from First Principles

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{B.32}$$

B.1.7 Numerical Solutions of Equations

The Newton-Raphson iteration for solving $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{B.33}$$

B.1.8 Numerical Integration

The Trapezium Rule:

$$\int_a^b y dx \approx \frac{1}{2} \left\{ h(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1}) \right\}, \text{ where } h = \frac{b-a}{n} \tag{B.34}$$

B.1.9 Further Numerical Integration

The mid-ordinate rule:

$$\int_a^b y dx \approx h \left(y_{\frac{1}{2}} + y_{\frac{3}{2}} + \cdots + y_{n-\frac{3}{2}} + y_{n-\frac{1}{2}} \right) \tag{B.35}$$

where $h = \frac{b-a}{n}$
Simpson's Rule:

$$\int_a^b y dx \approx \frac{1}{3} h \left\{ (y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1} + 2(y_2 + y_4 + \cdots + y_{n-2})) \right\} \tag{B.36}$$

B.1.10 Numerical Solutions to Differential Equations

For $\frac{dy}{dx} = f(x)$ and small h, recurrence relations are:
Euler's Method:

$$y_{n+1} = y_n + hf(x), \quad x_{n+1} = x_n + h \tag{B.37}$$

For $\frac{dy}{dx} = f(x, y)$:
Euler's Method:

$$y_{r+1} = y_r + hf(x_r, y_r), \quad x_{r+1} = x^r + h \tag{B.38}$$

Improved Euler's Method:

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r), \quad x_{r+1} = x_r + h \tag{B.39}$$

B.1.11 3D Matrices

The matrices for rotations (in three dimensions) though an angle θ about one of the axes are:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad \text{for the } x\text{-axis} \tag{B.40}$$

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \text{for the } y\text{-axis} \tag{B.41}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{for the } z\text{-axis} \tag{B.42}$$

B.2 Mechanics

B.2.1 Momentum and Impulse

$$\mathbf{p} = m\mathbf{v} \tag{B.43}$$

$$\mathbf{p} = \mathbf{F}t \tag{B.44}$$

$$\mathbf{I} = m\Delta\mathbf{v} = m(\mathbf{v} - \mathbf{u}) \tag{B.45}$$

$$\mathbf{I} = \Delta\mathbf{p} = m\mathbf{v} - m\mathbf{u} \tag{B.46}$$

B.3 Probability and Statistics

Glossary

algorithm A set of instructions instructing a person or computer on what steps to follow to achieve a goal. See section 3.1. 29, 30, 31, 38, 48, 131, 132, 133

ampere The SI unit of electric current. 132

Argand diagram A 2D graph of the complex plane. 5, 6, 8, 9, 10

argument The angle that a complex number makes with the positive real number line. It is written as $\arg(z)$. The argument is in the real interval $(-\pi, \pi]$. 5, 6, 7, 9, 10

bubble sort A sorting algorithm that looks through each pair of items and swaps them if they are in the wrong order. It has an order of $\mathcal{O}\left(n^2\right)$. 30, 31, 133

candela The SI unit of luminous intensity. It is approximately the luminous intensity of a wax candle. 132

Cartesian equation An equation relating points on the Cartesian plane, e.g. $y = x+1$. 8, 9, 10

Cartesian graph A graph in the x-y plane. 2, 5, 132

Cartesian plane The 2D plane of real points (x, y). 8, 129

complete graph A graph with an edge connecting every pair of vertices. A complete graph with n vertices is written K_n. 41, 42, 46

complex conjugate A complex number with a flipped sign before its imaginary part. The complex conjugate of a complex number z is written z^*. 2, 6, 7, 129

complex number A number in the form $a + bi$, where $a, b \in \mathbb{R}$. The set of complex numbers is written $\mathbb{C}$. 1, 2, 5, 6, 7, 129, 131, 132

complex plane The 2D plane of that the complex numbers are on. 8

complex root A root x of an equation where $x \in \mathbb{C}$. It is often also used to mean that $x \notin \mathbb{R}$. 2

conjugate pair A pair of complex numbers, where one is the complex conjugate of the other. 2

connected graph A graph where every vertex has an edge to or from every other vertex. 40, 46, 50, 133

cosine A trigonometric function. In a right-angled triangle with a hypotensuse H, opposite side O, adjacent side A and angle θ, $\cos\theta = \frac{A}{H}$. It can be related to the sine function with the identity $\cos\theta = \sin(\theta + \frac{\pi}{2})$. It is an odd function, meaning that $-\sin\theta = \sin(-\theta)$. Its definition is:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \tag{B.47}$$

Its reciprocal function is the secant function, sec. 6, 7

cubic A 3rd degree polynomial, i.e. an equation in the form $x^3 + bx^2 + cx + d = 0$, where $a \neq 0$. 2, 12, 13

cubic A 4th degree polynomial, i.e. an equation in the form $ax^4+bx^3+cx^2+dx+e = 0$, where $a \neq 0$. 12

cycle A walk where the first vertex visited is also the last vertex visited, and no other vertex is visited twice. 43, 46, 47, 48, 49, 50, 130, 133

degree (Graph theory) The the number of edges connecting to a given vertex. 39, 43

directed edge An edge that only connects its two vertices in one direction, and cannot be travelled in the other direction. 45, 50, 130

directed graph A graph containing one or more directed edges. 40, 45, 46, 50, 130

edge A line connecting two vertices, also called an arc. Edges can have weights (weighted edges) and directions (directed edges). These are used to make weighted graphs and directed graphs respectively. 39, 40, 41, 42, 43, 44, 45, 46, 48, 49, 50, 129, 130, 131, 133

force A push or pull on an object. It has the SI units newtons (N). See equations 10.7. 66

function A map between two set (from the function's range to its domain) that maps each item in one (its range) to an item in the other (its domain). The image of x under the function f is written $f(x)$. A function that maps elements of X to elements in Y is written $f : X \to Y$. 6, 7, 130, 133

graph A set of vertices connected by edges, also called a network. See section 3.2. 39, 40, 41, 43, 44, 45, 46, 47, 48, 49, 50, 51, 129, 130, 133

Hamiltonian cycle A cycle where every vertex the graph is visited. 43

imaginary constant The square root of -1, written i. 1

imaginary part The coefficient of the imaginary term of a complex number, i.e. where $z = a+bi,\ a, b \in \mathbb{R}$, the imaginary part of z is b. The imaginary part of a complex number z is written as $imag(z)$ or $\mathfrak{I}(z)$. 1, 5

integer A number with no decimal part. The set of all integers ($\{\cdots -3, -2, -1, 0, 1, 2, 3, \cdots\}$) is written as $\mathbb{Z}$. 29

kelvin The SI unit of temperature. A change of temperature of 1K is the same as a change of 1°C, but 0K is approximately -273°C. 132

kilogram The SI unit of mass. It appears to have an SI prefix (kilo), however due to historical reasons, it is considered a base unit. The gram still exists, defined as a thousandth of a kilogram. 65, 131, 132

Kruskal's algorithm An algorithm that finds a minimum spanning tree of a given weighted graph. It has an order of $\mathcal{O}(f(x))$. 48, 50

locus A set of points that satisfy some criteria, such as $\{z \in \mathbb{C} : |z| = 1\}$. 8, 9, 10

loop An edge that connects a vertex to itself. 40, 41, 42, 46, 133

mass The amount of matter in an object. Its SI unit is the kilogram (kg). 65, 131

metre The SI unit of length. 65, 131, 132

minimum spanning tree A spanning tree S_1 of a weighted graph G such that there does not exist another spanning tree S_2 of G where the sum of weights of S_2 is less than the sum of weights of S_1 and $S_1 \ncong S_2$. 47, 48, 51, 131

modulus The magnitude of a number, where $|x|$ is the modulus of x. For positive real numbers, the modulus is itself, for negative real numbers, the modulus is the positive version of itself ($|x| = -x,\ x \in \mathbb{R}^-$). For a complex number z, $|z| = \sqrt{\mathrm{Re}^2(z) + \mathrm{Im}^2(z)}$. 5, 6, 8

modulus-arguemnt form The form $r(\cos\theta + i\sin\theta) \equiv \mathrm{cis}\,\theta$. 5

mole The SI unit of quantity. 1mol is approximately the number of atoms of carbon-12 in 12g of carbon-12. It is defined as N_A^{-1}. 132

momentum The product of an object's mass and momentum. It the SI units kilogram metres per second ($\mathrm{kg\,m\,s^{-1}}$), or newton seconds (N s). See equations B.43, B.44. 65, 66

momentum Velocity is a vector quantity measuring the rate of change of position of an object. See equations 10.7. 65, 66, 131

newton The SI unit of force. The newton is not a base unit, but instead is equivalent to a kilogram metre per second squared ($\mathrm{kg\,m\,s^{-2}}$). 65, 130, 131

order The rate at which the time taken to complete an algorithm increases based on the size of its input. It is written as $\mathcal{O}(f(n))$, where f is some function, generally $f : \mathbb{N} \to \mathbb{R}$, and n is the size of the input. The order of an algorithm ignores anything excpet the greatest term of f, and discards constnat coefficients. 129, 131

path A walk where no vertex is visited twice. 43, 50

polynomial An expression in the form $a_0 + a_1x^1 + a_2x^2 + \cdots + a_nx^n$. 1, 2, 12, 13, 14, 15, 130, 132, 133

quadrant A quarter of a graph; the 1st quadrant is the top right, the second is the top left, the third is the bottom left, and the fourth is the bottom right.. 5, 6

quadratic A second degree polynomial, i.e. an equation in the form $ax^2 + bx + c = 0$, where $a \neq 0$. When plotted on a Cartesian graph, it forms a parabola. Its roots are given by the quadratic formula, $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$. 2, 12, 13, 15, 125

radian An angle unit, equal to the angle required to make the arc length of a unit circle equal 1. A whole circle is 2π radians; half a circle is π radians. The symbol for radians is either ^{c} or rad, as in π^{c} or π rad, or is often omitted. 5, 7, 77

real number A number on the real number line; a number in the interval $(-\infty, \infty)$. The set of real numbers is written $\mathbb{R}$. 1, 2, 131

real part The real term of a complex number, i.e. where $z = a + bi,\ a, b \in$=, the real part of z is a The real part of a complex number z is written as $\text{Re}(z)$ or $\Re(z)$. 1, 5

realising the denominator The process of multiplying a complex fraction in the form $\frac{w}{z}$ by $\frac{z^*}{z^*}$ (where $w, z \in \mathbb{C}$) to make the denominator a real value without changing the value of the fraction.. 2

reciprocal The reciprocal of a number is its multiplicative inverse, i.e. for some number x with a reciprocal x^{-1}, $x \times x^{-1} = 1$. 12, 13, 130, 133

root A value that satisfies an equation, for example 1 and -1 are the roots of the equation $x^2 - 1 = 0$, since for $x = 1$ and $x = -1$, the equation holds. 2, 12, 13, 14, 15, 116, 125, 132, 133

second The SI unit of time. 65, 131, 132, 133

SI unit The International System of units; the standard unit system used globally for maths and other STEM subjects. All SI units are made from the 7 base units: the metre (m), kilogram (kg), the second (s), the ampere (A), the kelvin (K), the mole (mol) and the candela (cd). All of the base units are derived from 7 universal constants: the speed of light (c), the hyperfine transition frequency of caesium-133, the Planck constant (h), the Avrogadro constant (N_A), the luminous efficacy

of 540THz radiation, the Boltzmann constant (k), and the elementary change (e). 65, 129, 130, 131, 132, 133

simple graph A graph with no more than one edge between any two vertices and no loop. 40, 41

sine A trigonometric function. In a right-angled triangle with a hypotensuse H, opposite side O, adjacent side A and angle θ, $\sin\theta = \frac{O}{H}$. It is an odd function, meaning that $-\sin\theta = \sin(-\theta)$. Its definition is:

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \tag{B.48}$$

Its reciprocal function is the cosecant function, csc. 6, 7, 130

sorting algorithm A type of algorithm that takes a list of items and sorts them. One example is bubble sort. 31, 129

spanning tree A spanning tree of a graph G is a tree that is a subgraph of G that contains all of the vertices in G. 47, 48, 49, 50, 51, 131

subgraph A graph $g = (v, e)$ is a subgraph of another graph $G = (V, E)$ if $v \subseteq V$ and $e \subseteq v^2 \subseteq E$. 43, 44, 47, 133

time The progression of existence, represented as the fourth dimension of (3+1)D spacetime. It has the SI units seconds (s). 66

trace table A table that tracks variables while an algorithm is running. 29

trail A walk where no edge is visited twice. 43

tree A connected graph with no cycles. 46, 47, 49, 50, 133

trigonometry The field of maths that studies trigonometric functions (sin, cos, tan, sinh, cosh, tanh, etc.). 6, 130, 133

vertex A point of a graph, also called a node. 39, 40, 41, 42, 43, 44, 46, 47, 48, 50, 129, 130, 131, 132, 133

Vieta's formulae A set of formulae relating the roots of a polynomial to its coefficients. 12, 13

walk A route along edges of a graph. 43, 130, 132, 133

weighted edge An edge with a weight assigned to it. This is a scalar usually representing a real-world value. For example, if each edge represents a road connecting two places (two vertices), then the weight could represent the length of the road, or the time taken to travel it. 44, 130, 133

weighted graph A graph containing one or more weighted edges. 44, 47, 48, 50, 130, 131